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isk-Annotated Spatial Scene Graph for Dynamic Risk Assessment in Autonomous Systems

Master Thesis TBD

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Abstract

Conventional risk assessment mechanisms are limited when the control strategy or operating environment of an autonomous system changes after deployment. This limitation is particularly relevant for systems operating in dynamic environments or modified through adaptive software components and over-the-air updates.

This thesis extends the framework for online risk-annotated scene graph generation presented by André Schiel in the master's thesis *Online Risk-Annotated Scene Graphs for Autonomous Systems* [1] into a spatial scene graph that supports dynamic risk assessment in near real time. The scene graph represents objects together with their spatial context, semantic labels and risk-relevant object descriptions generated by vision-language models, and an uncertainty estimate of whether each object is still at its last observed position. The framework is architected as a modular ROS 2 system, which supports portability and integration with robotic platforms, and it builds on the Hydra spatial perception system to support large environments.

Development and optimisation were carried out on an NVIDIA Jetson AGX Orin using the office scene of the simulated uHumans2 dataset, and the system was subsequently ported to an NVIDIA Jetson AGX Thor. Qwen3.5-4B combined with an optimised prompt was selected for object labelling, reaching an accuracy of 88 % on 50 manually verified object crops. Additional tests on the uHumans2 apartment scene and on multiple recordings from the office scene of the real-world OpenLORIS-Scene dataset demonstrate portability across environments and across simulated and real sensor data. On the Jetson AGX Thor, the pipeline processes **[XX]** frames per second on **[dataset and configuration]**. This near-real-time rate is an improvement over the baseline framework and makes the risk-annotated scene graph available as environmental context during system operation.

Keywords: dynamic risk assessment, risk-annotated scene graph, ROS 2, Hydra, uncertainty-aware object representation, vision-language model, edge AI

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1. Introduction

1.1. Motivation

Robotic systems are increasingly moving out of fenced industrial cells and into shared, dynamic environments such as warehouses, hospitals, offices, and public spaces. According to the International Federation of Robotics, almost 200,000 professional service robots were sold in 2024, an increase of 9% over the previous year [2]. The largest share of these, 102,900 units, were built for transportation and logistics and consist mainly of mobile robots that move goods through their surroundings [2]. At the same time, the global operational stock of industrial robots reached 4.66 million units in 2024 [3]. Together, these figures show that robots are becoming a common part of environments that are also used by people.

Alongside this growth, robotic systems are increasingly shaped by artificial intelligence. The International Federation of Robotics identifies physical, analytic, and generative AI as key technological trends in robotics [3]. AI-based components allow robots to perceive their surroundings, interpret scenes, and adapt their behaviour. At the same time, they make robot behaviour harder to predict fully during design-time safety analysis [4].

This development creates new challenges for safety and risk assessment. Conventional risk assessment is typically carried out before deployment and relies on assumptions about the system design, intended use, operating environment, and known hazards [5]. These methods remain essential, but they reach their limits when mobile robots operate in changing environments, receive over-the-air software updates, or use adaptive AI-based components. In such cases, the situation the robot actually faces may differ from the one assumed during the original assessment. For this reason, automated and continuous risk assessment has been proposed, in which the assessment is updated while the system is running [6].

Such dynamic risk assessment needs a structured description of the current environment. It must know which objects are present, where they are, how they relate to each other, and which of them may be relevant to risk. A spatial risk-annotated scene graph is a suitable way to provide this information, as it combines perceived objects, their spatial context, and risk-related properties in a single machine-readable representation. In this thesis, the term *risk-annotated scene graph* refers to a spatial scene graph whose object nodes are enriched with risk-relevant semantic and uncertainty information, including object labels, confidence values, mobility information, presence likelihood, and descriptive object properties. These annotations provide structured environmental context for downstream dynamic risk assessment rather than representing a complete risk assessment by themselves.

1.2. Relevance

Scene graphs have become an established way to represent robotic environments in a structured form. Hydra [7] builds a layered 3D scene graph of objects, places, and rooms in real time and keeps it consistent through loop closure. However, Hydra does not identify objects itself. Its object layer depends on per-pixel semantic labels, which come from ground-truth annotations in simulation or from a segmentation network trained on a fixed set of classes [7]. ConceptGraphs [8] removes this restriction by identifying objects with open-vocabulary foundation models, but it applies these models to every observation and represents objects through free-form descriptions and features rather than bounded, confidence-weighted information.

At the same time, work on safety reasoning shows that scene graphs are a useful input for detecting hazardous situations. SafetyDetect [9], for example, uses object relations from a scene graph to let a language model classify situations as normal or dangerous. The authors note that their approach is limited mainly by how the scene graph is created, since risk assessment can only consider what the scene graph actually contains. The quality of the underlying scene graph is therefore a key factor for any risk assessment built on top of it.

The existing risk-annotated scene graph framework [1] brings these two directions together by adding risk-related information to a scene graph generated from RGB-D data. It is, however, limited to local, frame-based scene understanding and is exposed through a service interface that is not designed for robotic middleware. It also has no mechanism for keeping objects consistent across a larger environment, and it provides only basic handling of perception uncertainty. There is therefore a need for a risk-annotated scene graph that combines the spatial consistency of systems like Hydra with flexible object identification and risk-relevant object information, and that is designed for integration into a mobile robotic system during operation.

1.3. Goal of the Thesis

The goal of this thesis is to extend the existing risk-annotated scene graph framework [1] into a 3D spatial scene graph that provides up-to-date environmental context for dynamic risk assessment. The scene graph should describe the objects in the environment, their spatial context, and their risk-relevant properties. The work focuses on four aspects: portability, spatial scalability, uncertainty awareness, and near-real-time operation.

The first goal is to integrate the baseline framework into ROS 2 [10], so that it can interact directly with robotic sensors, localisation, and mapping components. The second goal is to ground the scene graph in a spatial map, so that objects observed at different times and locations share a common reference frame and remain spatially consistent when areas are revisited. Object identification should support open-vocabulary semantic perception, removing the dependence on ground-truth semantic labels or a fixed set of object classes.

The third goal is to represent uncertainty associated with object identification, mobility, and continued presence in the environment. The fourth goal is to support near-real-time operation on embedded robotic hardware. Finally, the framework is evaluated on the simulated uHumans2 dataset [11] and the real-world OpenLORIS-Scene dataset [12], focusing on object tracking and labelling, scene graph consistency, and runtime behaviour.

1.4. Contributions

This thesis makes the following contributions:

1. **A ROS 2-based risk-annotated scene graph framework.** The baseline framework is adapted for ROS 2, enabling direct integration with the sensors and software components of a robotic system. This provides a foundation for using risk-relevant scene representations within a broader robotic software architecture.

2. **A spatial scene graph with open-vocabulary object identification.** The framework extends Hydra [7] with flexible object identification beyond fixed semantic classes. The resulting 3D scene graph combines spatial, semantic, and mobility information while maintaining a consistent representation of the environment over time.

3. **A persistent object representation with uncertainty information.** Objects are represented together with confidence information for their semantic identity, mobility, and continued presence in the environment. This allows the scene graph to express both what is currently believed about an object and the uncertainty associated with that information.

4. **A design for near-real-time operation on embedded robotic hardware.** The framework is designed to maintain spatial scene understanding while incorporating computationally demanding semantic information. This allows the scene representation to remain suitable for applications in which environmental information must be updated continuously during robot operation.

5. **An experimental evaluation on simulated and real-world data.** The extended framework is evaluated using the simulated uHumans2 dataset [11] and the real-world OpenLORIS-Scene dataset [12]. The evaluation examines object tracking and labelling, spatial scene graph consistency, and runtime behaviour to identify both the capabilities and limitations of the proposed approach.

2. Theoretical Background

2.1. Autonomous Systems and Safety

Autonomous systems (AS) can be understood as systems that are capable of making decisions independently from direct human control [13]. In robotic systems, this autonomy is typically achieved through a combination of sensing, perception, planning, decision-making, and actuation. For example, an autonomous mobile robot may perceive obstacles, select an alternative path, and continue its task without requiring continuous human input. This ability allows autonomous robots to operate more flexibly in dynamic and partially unknown environments. However, it also makes safety assurance more complex, because the system behaviour is no longer determined only by predefined commands, but also by runtime perception, environmental conditions, and decision-making processes.

Safety is a fundamental requirement in AS because robots interact with humans, physical objects, infrastructure, and the surrounding environment. In dependability theory, safety is commonly defined as the absence of catastrophic consequences for users and the environment [14]. In the context of autonomous robotic systems, safety therefore refers not only to whether the robot performs its intended task, but also to whether it performs the task without creating unacceptable risk. This distinction is important because a robot may be functionally successful while still being unsafe. For example, a robot arm in a shared workspace may complete a pick-and-place task correctly, but the system cannot be considered safe if its motion exposes a nearby human operator to an unacceptable collision risk.

From a risk assessment perspective, safety is closely related to the identification of hazards, the estimation of risk, and the implementation of suitable risk reduction measures [5]. In robotic applications, this requires considering both the severity of possible harm and the likelihood that a hazardous situation may occur. Robot-specific safety standards further define requirements and protective measures for industrial and collaborative robotic systems [15]–[17]. However, safety in AS is strongly dependent on operational context. A motion, speed, or decision that is safe in one situation may become hazardous when a human enters the workspace, when an object is misplaced, or when perception uncertainty increases. Therefore, safety in autonomous robotic systems must be understood as a property that depends on both the technical system and the changing environment in which it operates.

2.1.1. Human-Robot Collaboration

Industry 5.0 is shifting industrial development toward a more human-centric, sustainable, and resilient model, while continuing to support productivity and worker well-being [18]. Within this perspective, Human-Robot Collaboration (HRC) becomes an important requirement for using robotic technologies to support and empower workers rather than simply replacing human labour [18]. Collaborative robotic systems can assist humans in repetitive, strenuous, or hazardous tasks, thereby improving workplace flexibility, ergonomics, and the distribution of work between humans and machines. In industrial settings, HRC has also been discussed as a response to skill gaps, skilled-worker retention, and the need to make manufacturing work more attractive to younger generations [19]. This makes HRC a relevant foundation for future autonomous systems, where robots increasingly need to interpret human presence, environmental changes, and task-specific context during operation.

However, HRC also introduces important challenges in industrial settings. Industrial HRC must address three closely related aspects: ensuring human safety, maintaining trust in automation, and preserving productivity [19]. Among these, human safety remains a primary concern because close physical interaction increases the possibility of unintended human-robot collisions, unexpected robot motion, or interruptions in the robot's behaviour. Such situations can increase the risk of injury and may negatively affect the human operator's trust in the system. At the same time, safety measures must be balanced with productivity requirements, since overly restrictive measures such as excessive speed reductions, large separation distances, or frequent safety stops can reduce system efficiency and interrupt the intended collaborative workflow [19]. The relationship between safety, trust in automation, and productivity is illustrated in Figure 2.1. This trade-off shows that HRC cannot be addressed only as a productivity or automation problem; it also requires systematic risk assessment methods that can identify hazardous situations and define suitable risk reduction measures.

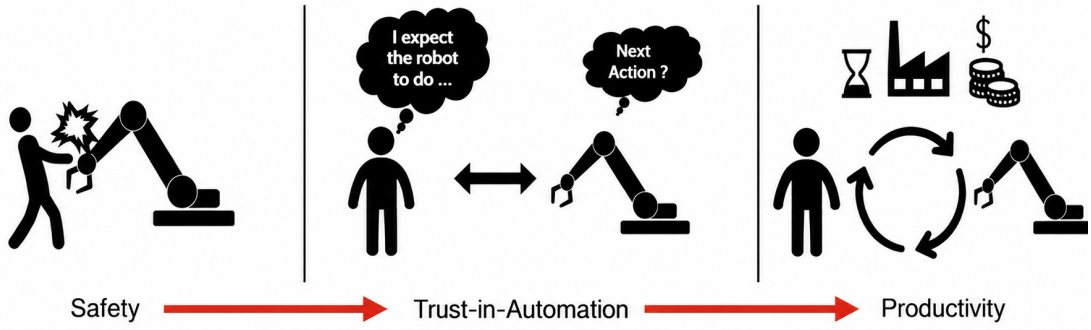


Figure 2.1.: Main challenges in HRC[19].

2.1.2. Conventional Risk Assessment and Risk Reduction

Conventional risk assessment in robotic systems is generally performed before deployment and is based on the intended use, foreseeable misuse, system limits, operating environment, and known hazards of the system. According to ISO 12100, the risk assessment process includes hazard identification, risk estimation, and risk evaluation, followed by the selection of suitable risk reduction measures [5]. In the context of industrial and collaborative robots, this process is supported by robot-specific safety standards that define requirements for robot systems, collaborative applications, and protective measures [15]–[17].

Risk reduction is usually implemented through a combination of inherently safe design, technical protective measures, and information for use. In HRC, this can include limiting robot speed and force, defining safe separation distances, using protective stops, and applying physical or electronic safeguards around the robot workspace. These measures are essential because they provide a systematic way to reduce the likelihood or severity of hazardous human-robot interactions before the system is put into operation.

However, conventional risk assessment is mainly based on assumptions made during design and deployment. These assumptions include expected human behaviour, known object configurations, defined workspaces, and predefined task conditions. As a result, conventional risk assessment is effective for identifying known and reasonably foreseeable hazards, but it becomes limited when the operating context changes significantly during runtime. This limitation is especially relevant for autonomous robotic systems, where the robot may operate in dynamic environments and make decisions based on sensor data and perception results.

2.1.3. Existing Safety Measures in HRC

In industrial HRC, risk reduction strategies are commonly implemented through two main approaches: the use of collaborative robots and the use of physical or electronic safeguarding measures [19]. Collaborative robots, also referred to as cobots, are designed to operate in direct cooperation with humans within a defined shared workspace. Their safety is supported through hardware and software features such as reduced momentum, force sensing, back-drivable actuators, and protective stop functions that interrupt robot motion when unexpected contact or resistance is detected.

The safeguarding-based approach focuses on monitoring the robot workspace so that robot motion can be adapted or stopped before hazardous human-robot collisions occur. In practice, this is achieved using physical and electronic safeguarding devices that define protected zones around the robot work cell. Examples include safeguarded perimeter door switches, light curtains, pressure-sensitive floors, 2-D scanning LiDAR, and 3-D safety vision systems. These mechanisms are important for reducing the risk of human injury, but they often rely on predefined safety zones, fixed distance thresholds, or conservative stopping rules [19].

A common method used for workspace monitoring is Speed and Separation Monitoring (SSM). In SSM, the distance between the human and the robot is monitored continuously, and the robot speed is reduced or stopped when the separation distance becomes too small. In a static SSM approach, the monitored zones are usually predefined before operation. For example, a robot may operate at full speed when no human is detected in the outer zone, slow down when a human enters a warning zone, and stop when the human enters a protective stop zone[19]. This approach is simple and practical, but it can be conservative because the safety zones are not always adapted to the actual robot motion, human motion, task state, or sensor uncertainty.



Figure 2.2.: Illustrative representation of safety zones in a collaborative robot workspace.

Dynamic SSM addresses this limitation by adapting the protective separation distance or the robot speed according to the current operating situation. Instead of relying only on fixed zones, dynamic SSM considers factors such as the actual distance between the human and the robot, robot velocity, human motion, stopping time, and uncertainty in the sensed position of obstacles or operators. Marvel discusses SSM performance in terms of both safety and productivity, representing collision potential through separation metrics and sensor uncertainty [20]. This shows that SSM is not only a binary stop-or-run mechanism, but also a method whose effectiveness depends on how accurately the system evaluates separation, uncertainty, and the productivity cost of safety actions.

Although dynamic SSM provides a more flexible basis for HRC safety than purely static safety zones, it also requires reliable perception, continuous monitoring, and suitable runtime decision-making. Therefore, existing safety measures show the importance of moving from fixed protective

assumptions toward context-aware risk assessment mechanisms that can consider the actual operational situation of the robot, humans, and surrounding environment.

2.1.4. Impact of Artificial Intelligence

The previous subsections show that conventional risk assessment and many existing HRC safety measures are based on pre-deployment assumptions, such as intended use, expected human behaviour, known object configurations, predefined workspaces, fixed safety zones, and foreseeable hazardous scenarios. These assumptions are necessary for systematic risk assessment, but they become less reliable when autonomous robots operate in changing environments. Human positions, object locations, obstacles, task conditions, and spatial relationships may vary during operation, meaning that a risk analysis performed before deployment may no longer fully represent the actual operating situation.

The integration of artificial intelligence into robotic perception, planning, or control further increases this limitation. AI-based components can enable object detection, scene interpretation, motion prediction, task planning, and adaptive decision-making. While these capabilities improve flexibility and autonomy, they also make robot behaviour harder to predict completely during design-time safety analysis. Safety assurance of AI-based systems is complicated by dataset dependence, black-box behaviour, explainability, and lifecycle assurance [4]. In robotic systems, this means that perception errors, low-confidence predictions, distribution shifts, previously unseen objects, changing human behaviour, or software updates after deployment can affect system decisions.

As a result, conventional risk assessment should be understood as a necessary baseline rather than a complete solution for adaptive autonomous robotic systems. If the deployed system operates under conditions that were not fully represented during the original assessment, the selected safety measures may become insufficient. Therefore, dynamic and context-aware risk assessment is required to incorporate runtime information such as robot state, human presence, object locations, spatial relationships, perception uncertainty, and environmental changes. Automated and continuous risk assessment has been proposed for software-defined robotic systems, where changes in software or system behaviour may require risk assessment to be updated during operation [6].

At the same time, AI can also support dynamic risk assessment when its outputs are structured, bounded, and interpretable. AI-based perception and reasoning methods can help identify relevant objects, detect environmental changes, classify scene elements, and localise risk-relevant information around the robot. In this thesis, this motivates the use of a spatial risk-annotated scene graph, which can represent objects, spatial relations, uncertainty, and risk-related context in a machine-readable form to support dynamic risk assessment in autonomous robotic systems.

2.2. Existing Work on Spatial Scene Graphs

Spatial scene graphs extend image-level scene graph representations by grounding objects, relationships, and higher-level spatial concepts in 3D space. This is important for robotic systems because a robot must not only recognise objects, but also understand where they are located, how they are arranged, and how they relate to larger spatial structures such as rooms or buildings. Several previous works have proposed methods for generating 3D scene graphs from reconstructed environments, RGB-D observations, or open-vocabulary perception pipelines. Although these works do not directly focus on risk annotation, they provide important methodological foundations for the spatial scene graph generation problem addressed in this thesis.

2.2.1. 3D Scene Graph by Armeni et al.

Armeni et al. introduced one of the foundational formulations of 3D scene graphs for building-scale indoor scene understanding [21]. Their work extends the conventional scene graph concept from 2D image representations to 3D environments. Instead of representing only objects and pairwise relations in an image, the proposed 3D scene graph organises semantic and spatial information across multiple levels of abstraction, including buildings, rooms, objects, and cameras.

The method uses a reconstructed 3D mesh and registered panoramic images as input. Visual information from the images is associated with the 3D mesh, allowing semantic labels and object-level information to be grounded in physical space. The resulting graph represents not only which objects are present, but also where they are located and how they are connected to larger spatial entities such as rooms. This makes the representation more suitable for indoor spatial understanding than a purely image-level graph.

A key contribution of this work is the introduction of a hierarchical representation for complete indoor environments. Object nodes can be connected to room nodes, cameras can be connected to the regions they observe, and spatial relationships can be represented explicitly. This supports queries such as which objects are present in a room, how cameras observe the environment, or how semantic information is distributed across a building.

For the present thesis, Armeni et al. are mainly relevant as a conceptual foundation for spatial scene graph structure. Their work shows how semantic information can be organised across object, room, camera, and building levels. However, the method is primarily designed for building-scale scene understanding from reconstructed data, rather than online robotic operation. Therefore, its main contribution to this thesis is the idea of spatial hierarchy, while real-time construction, loop closure, uncertainty handling, and risk annotation require additional methods.

2.2.2. SceneGraphFusion

SceneGraphFusion by Wu et al. focuses on incremental 3D scene graph prediction from RGB-D sequences [22]. This makes it particularly relevant to robotic perception, because RGB-D cameras provide both colour and depth information and are commonly used for indoor robotic systems. Unlike methods that generate a scene graph from a single image or from a completed offline reconstruction, SceneGraphFusion builds and updates the graph incrementally as new RGB-D frames are observed.

The method first uses RGB-D input to support 3D reconstruction and instance-level scene understanding. As the camera observes the environment from different viewpoints, partial observations are fused into a growing 3D representation. Object instances are represented as graph nodes, while relationships between objects are represented as graph edges. The method then predicts object and relationship semantics using graph neural network-based reasoning, combining geometric, visual, and semantic information from the reconstructed scene.

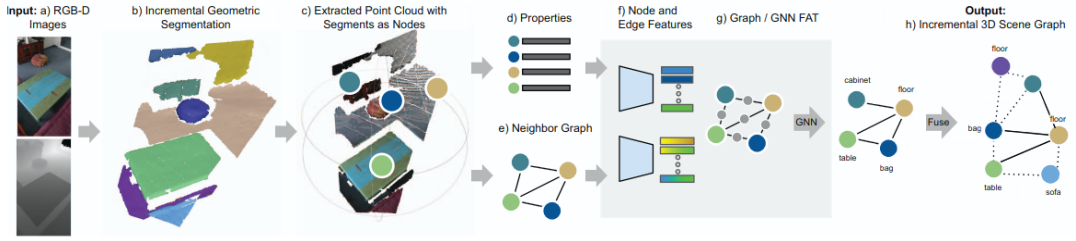


Figure 2.3.: Overview of the SceneGraphFusion pipeline, showing RGB-D input, incremental geometric segmentation, point-cloud segments as graph nodes, feature extraction, graph neural network-based relationship prediction, and temporal fusion into an incremental 3D scene graph. Reproduced from [22].

A key contribution of SceneGraphFusion is its incremental formulation. Robotic perception is naturally sequential: a robot observes only part of the environment at each time step, and the representation must be updated as new views become available. SceneGraphFusion addresses this problem by treating scene graph generation as a progressive RGB-D fusion task rather than as a single-frame prediction problem. This allows the graph to become more complete as additional observations are integrated.

An important part of the method is its temporal scene graph fusion strategy. Because the framework predicts the semantic labels of segments and relationships multiple times over a sequence, different observations may produce different or contradictory predictions. To handle this, SceneGraphFusion stores a probability distribution and a weight for each predicted class or relationship predicate. When a new prediction is available, the stored probability is updated using a weighted running average. Using $\hat{\mu}_t$ for the new prediction probability and $\hat{w}_t$ for its weight, the updated probability can be written as

$$\mu_t = \frac{\hat{\mu}_t \hat{w}_t + \mu_{t-1} w_{t-1}}{\hat{w}_t + w_{t-1}},$$

and the corresponding weight is updated as

$$w_t = \min(w_{\max}, \hat{w}_t + w_{t-1}),$$

where $w_{\max}$ is the maximum allowed weight [22]. This temporal fusion mechanism allows repeated observations to strengthen stable predictions while reducing the influence of isolated inconsistent predictions.

The temporal fusion strategy is relevant to this thesis as an example of how repeated observations can be combined to improve the stability of a scene graph over time. Instead of storing only a single deterministic label for each object or relationship, SceneGraphFusion preserves a probability distribution over possible classes or predicates. This allows the representation to retain information about semantic uncertainty and to reduce the effect of isolated inconsistent predictions. Such an approach is useful to study because dynamic risk assessment may depend not only on the selected object or relation label, but also on how confident the system is in that prediction.

For this thesis, SceneGraphFusion is therefore considered as an important reference for alternative approaches to incremental scene graph generation. It demonstrates how RGB-D observations can be fused over time and how semantic predictions for graph nodes and edges can be stabilized using temporal probability fusion. However, its main focus remains object-relation prediction in reconstructed indoor scenes. The present thesis investigates a different objective: extending risk-annotated scene graph generation toward spatial grounding, long-term object persistence, loop-closure-aware consistency, uncertainty representation, and dynamic risk assessment in robotic environments.

2.2.3. ConceptGraphs

ConceptGraphs by Gu et al. introduces open-vocabulary 3D scene graphs for perception and planning [8]. The work is motivated by the need for 3D scene representations that can be queried using flexible language-based concepts, rather than only a fixed set of predefined labels. This is relevant for robotic applications because the objects and categories encountered during operation may not always be known in advance. Instead of treating scene understanding only as a closed-set classification problem, ConceptGraphs connects 3D spatial mapping with language-aligned visual representations.

The method builds 3D scene graphs from posed RGB-D observations. Each RGB-D frame provides colour information, depth information, and a known camera pose. Object-level information is first extracted from individual 2D views using foundation-model-based perception. These detections

or segmentations provide candidate object regions in the image. The depth image and camera pose are then used to lift these 2D observations into 3D space. In this way, image-level object information is transformed into spatially grounded object candidates.

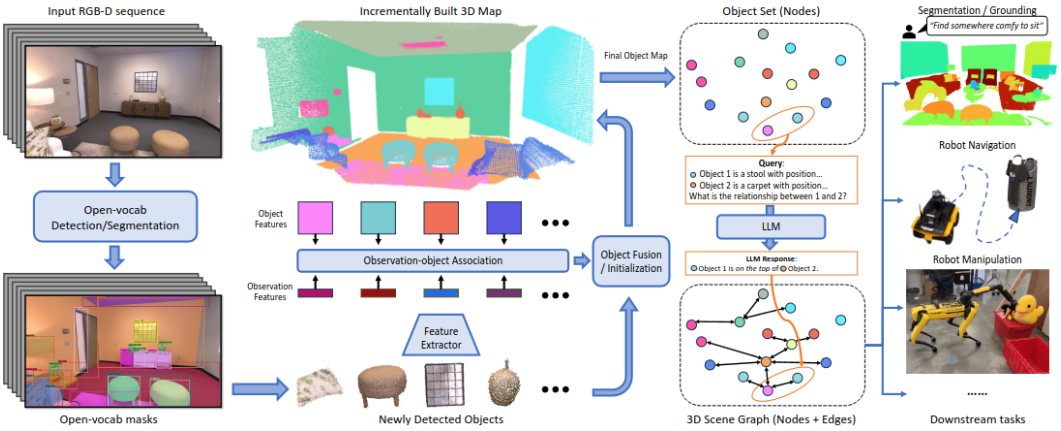


Figure 2.4.: Overview of the ConceptGraphs pipeline, showing open-vocabulary detection and segmentation from posed RGB-D observations, multi-view object association, object fusion, construction of a 3D scene graph, and downstream applications such as segmentation, grounding, navigation, and manipulation. Reproduced from [8].

A central part of the method is multi-view association. Since the same physical object can be visible in several RGB-D frames, the system must decide whether a newly detected object corresponds to an existing object in the 3D map or whether it should be inserted as a new object. ConceptGraphs performs this association using spatial and visual information, so that repeated observations of the same object can be fused into a single 3D object node. This reduces duplicate object creation and allows the graph to become more consistent as more views are processed.

The resulting graph contains object nodes that are grounded in 3D space while also carrying language-aligned features. These features allow the graph to be queried using natural-language descriptions. For example, instead of searching only for a fixed class label, a downstream module can use more flexible text queries related to object identity, appearance, or function. This makes the representation useful for tasks such as object grounding, navigation, manipulation, and language-conditioned planning.

For this thesis, ConceptGraphs is relevant because it shows one approach for combining 3D scene graph generation with foundation-model-based semantic understanding. This is related to the proposed risk-annotated scene graph framework, where visual-language reasoning is also used to enrich the scene representation. However, the focus of ConceptGraphs is mainly open-vocabulary perception and planning, whereas the present thesis focuses on generating a spatial scene graph that can support risk annotation, uncertainty representation, and dynamic risk assessment. Therefore, ConceptGraphs is useful as a reference for semantic flexibility in 3D

scene graphs, while the risk-oriented interpretation and safety relevance remain the focus of the proposed work.

2.2.4. OpenGraph

OpenGraph by Deng et al. proposes an open-vocabulary hierarchical 3D graph representation for large-scale outdoor environments [23]. While ConceptGraphs focuses on open-vocabulary 3D scene graphs from posed RGB-D observations, OpenGraph addresses outdoor scenes where the environment is larger, less structured, and contains many object instances. This makes the work relevant as an example of how open-vocabulary graph representations can be extended beyond indoor rooms or small-scale manipulation settings.

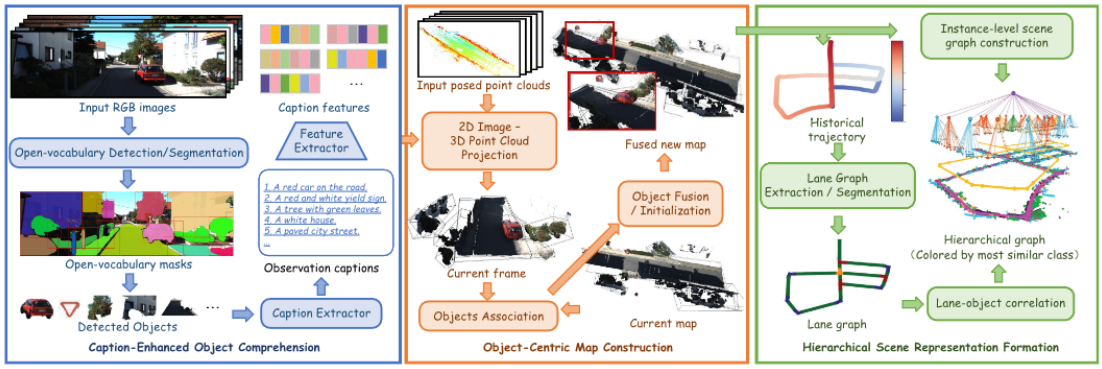


Figure 2.5.: Overview of the OpenGraph pipeline, showing caption-enhanced object comprehension from RGB images, object-centric 3D map construction using posed point clouds, and hierarchical scene representation formation for large-scale outdoor environments. Reproduced from [23].

The method first extracts object instances and textual captions from visual images. These captions are then encoded to support language-based reasoning about the detected instances. The 2D information is connected to 3D space by projecting image-based instance information onto LiDAR point clouds using multi-sensor calibration. In this way, OpenGraph builds an incremental object-centric 3D map in which objects are associated with both geometric information and language-related features.

A key feature of OpenGraph is its hierarchical organisation. The representation is not limited to isolated object nodes. Instead, the environment is structured using a hierarchy that includes object-level information and larger spatial regions derived from lane graph connectivity. This is useful in large outdoor environments, where direct object-level retrieval alone may be insufficient for navigation or spatial querying. The hierarchical graph structure allows the system to represent both local object information and broader environmental organisation.

For this thesis, OpenGraph is relevant because it shows how open-vocabulary semantic information can be combined with 3D graph representations in a large-scale robotic setting. Although the target environment is outdoor and the sensing setup relies on camera-LiDAR data rather than RGB-D input, the underlying idea is closely related: semantic information from visual-language models is grounded in 3D space and organised as a graph. This supports the motivation for using graph-based spatial representations when semantic reasoning must be connected to physical environments.

At the same time, OpenGraph has a different focus from the present thesis. Its main objective is open-vocabulary understanding, object retrieval, and hierarchical representation in outdoor environments. The present thesis instead focuses on spatial risk-annotated scene graph generation for robotic risk assessment. Therefore, OpenGraph is useful as a reference for large-scale and open-vocabulary graph construction, while the risk-related interpretation, uncertainty handling, and indoor RGB-D robotic integration remain separate contributions of this work.

2.3. Robot Operating System

The Robot Operating System (ROS) is an open-source software framework for developing and integrating robotic systems. Despite its name, ROS is not a traditional operating system; it functions as a middleware layer that provides communication mechanisms, software libraries, development tools, and conventions for building modular robot software [24]. In this thesis, ROS is relevant because the risk-annotated scene graph framework must be integrated with continuous RGB-D sensor streams, pose information, SLAM modules, system status information, and downstream risk assessment components. Therefore, ROS provides the conceptual and technical basis for replacing the service-oriented baseline interface with a robot-native integration approach.

2.3.1. ROS as Robotic Middleware

A ROS-based robotic system is typically organised as a collection of nodes. A node is an executable software component that performs a specific function, such as publishing camera data, processing sensor input, estimating robot pose, performing SLAM, planning motion, controlling actuators, or visualising system state. This node-based structure allows complex robotic systems to be decomposed into smaller modules that can be developed, tested, and replaced independently. For example, an RGB-D camera node can publish sensor data, a SLAM node can estimate the robot pose and map, and a scene graph node can use this information to update a structured representation of the environment.

Communication between nodes is performed using messages. A message defines the structure of the exchanged data, such as an image, pose, velocity command, point cloud, or status value. ROS provides standard message types for common robotic data, while also allowing custom message

definitions for application-specific information. This is important for robotic integration because different modules can exchange data in a predictable and structured way.

2.3.2. ROS Communication Mechanisms

ROS provides several communication mechanisms for different interaction patterns. Topics are used for continuous data streams and follow a publish-subscribe model. A node publishes messages to a topic, and one or more other nodes subscribe to that topic. This is suitable for sensor data such as RGB images, depth images, LiDAR scans, odometry, IMU data, and system status. In the context of this thesis, RGB-D observations, pose information, and scene graph updates can naturally be represented as topic-based communication because they are produced repeatedly during robot operation.

Services provide request-response communication. A service is useful when one node needs to request a specific operation from another node and receive a result, such as resetting a scene graph, saving the current graph, or requesting the current system state. Unlike topics, services are not intended for continuous streaming, but for discrete interactions where a response is expected.

Actions are used for long-running tasks that may require feedback or cancellation. An action consists of a goal, feedback during execution, and a final result. This mechanism is commonly used for navigation tasks, where a robot receives a goal pose, reports progress, and eventually returns whether the goal was reached. In a scene graph framework, actions could be useful for longer operations such as processing a selected dataset sequence or generating a scene graph for a defined exploration segment.

ROS also uses parameters to configure node behaviour. Parameters can define topic names, thresholds, model settings, update rates, frame identifiers, or other runtime options. This allows the same node to be reused in different experimental setups without changing the source code. Launch files are then used to start multiple nodes together with their required parameters, making it easier to reproduce complete robotic system configurations.

2.3.3. ROS 1 and ROS 2

ROS 1 was the first widely adopted version of ROS and became an important standard for robotics research and prototyping. In ROS 1, node discovery is coordinated through a central ROS Master, which allows publishers and subscribers to find each other before communicating. This architecture supported rapid development and experimentation, but it was mainly designed for research-oriented systems and did not fully address several requirements of modern robotic deployment, such as reliable distributed execution, multi-robot communication, real-time behaviour, security, and production-oriented system design.

ROS 2 was developed to address these limitations and to support more robust robotic applications [10]. A major architectural difference is that ROS 2 uses the Data Distribution Service (DDS) as its communication middleware. DDS enables decentralised discovery, configurable Quality of Service (QoS), and improved support for distributed robotic systems. This means that ROS 2 nodes can discover each other without relying on a central master, which improves suitability for multi-robot systems and distributed deployment.

Quality of Service is an important feature of ROS 2 communication. QoS settings allow developers to control how messages are delivered, for example whether communication should prioritize reliability or low latency. This is important because different robotic data streams have different requirements. A camera stream may prefer recent data and tolerate dropped frames, while system status or safety-related information may require more reliable delivery. ROS 2 also improves support for lifecycle management, embedded platforms, real-time control, and security compared with ROS 1 [10]. At the same time, practical studies of ROS 2 robotic control show that latency and jitter still need to be evaluated carefully in distributed and time-sensitive applications [25].

In the context of this thesis, ROS 2 is relevant because the risk-annotated scene graph framework must operate as part of a robotic system rather than as an isolated external service. RGB-D data, pose information, system status, and scene graph updates need to be exchanged continuously between different software components. ROS 2 provides a robot-native communication structure for this purpose, allowing the framework to be integrated with perception sensors, SLAM modules, robot state estimation, visualization tools, and downstream risk assessment mechanisms.

2.4. Scene Graphs

A scene graph is a structured representation of a visual scene in which objects are represented as nodes and relationships between objects are represented as edges [26]. Instead of describing an image or environment only as a collection of detected objects, a scene graph also captures how these objects are related to each other. For example, a scene may contain nodes such as *person*, *chair*, and *table*, while edges may describe relations such as *sitting on*, *next to*, or *under*. Johnson et al. demonstrated the usefulness of scene graphs for image retrieval by representing image content through objects, attributes, and relationships rather than relying only on low-level image features or isolated object labels [26].

The main advantage of a scene graph is that it provides a machine-readable representation of both entities and context. Object detections alone can indicate which objects are present, but they do not fully describe the structure of the scene. In contrast, a scene graph can represent object attributes, spatial arrangements, and semantic relationships in a compact graph format [26]. This makes it useful for tasks that require reasoning about the environment, because downstream modules can query not only whether an object exists, but also how it is positioned or related to other objects.

Scene graphs have been used in several computer vision and language-vision tasks, including image retrieval, visual relationship understanding, visual question answering, and image cap-

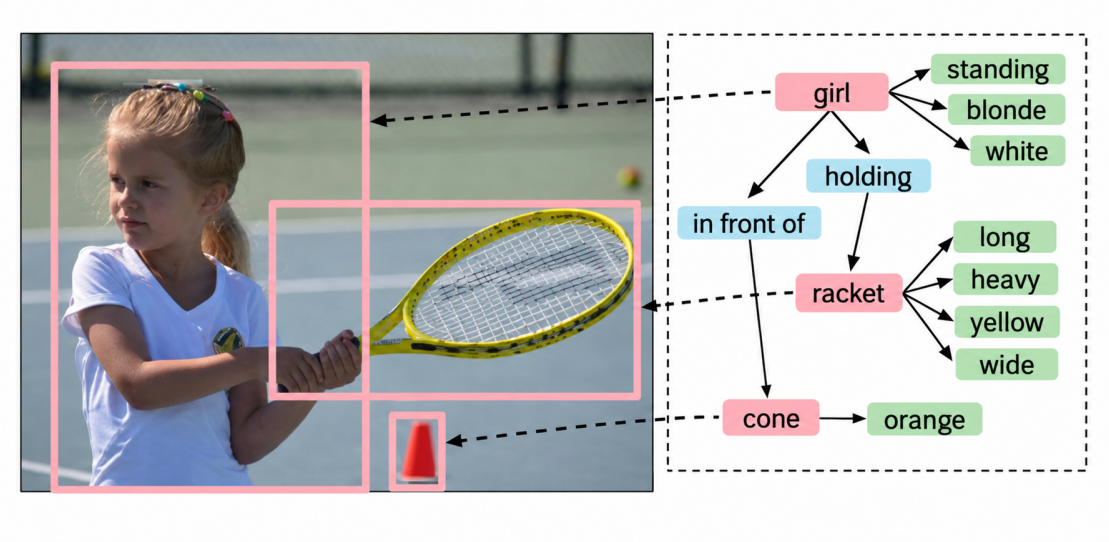


Figure 2.6.: Scene graph example. Reproduced from [26].

tioning. Datasets such as Visual Genome support this direction by providing dense annotations of objects, attributes, relationships, region descriptions, scene graphs, and question-answer pairs [27]. Beyond 2D images, scene graphs have also been extended to 3D environments, where objects, rooms, cameras, and their relationships can be represented in a unified spatial structure [21]. This makes scene graphs relevant not only for image-level understanding, but also for robotic and spatial perception tasks.

Scene graphs are also advantageous because they provide an interpretable bridge between perception and higher-level reasoning. Raw sensor data such as images or point clouds is difficult for symbolic reasoning modules to use directly. A scene graph abstracts this raw data into nodes, edges, and attributes that can be inspected, updated, and processed algorithmically. This is especially useful in robotic systems, where perception results must often be connected to planning, navigation, manipulation, or risk assessment.

In the context of this thesis, scene graphs are relevant because dynamic risk assessment requires more than object detection. Risk may depend on object identity, spatial proximity, support relations, containment, occlusion, human presence, and the current configuration of the environment. A graph-based representation allows such information to be stored in a structured form. When extended with spatial information, uncertainty estimates, and risk annotations, the scene graph becomes a suitable basis for representing the current operational context of an autonomous robotic system.

2.5. Hydra-Based Spatial Scene Graph Generation

Hydra is a real-time spatial perception system for constructing and optimizing 3D scene graphs from robotic sensor data [7]. It is relevant to this thesis because it addresses key requirements for spatial risk-annotated scene graph generation, including incremental construction, scalability to larger environments, loop closure correction, and near-real-time operation. While the baseline risk-annotated scene graph framework focuses on semantic and risk-related interpretation of RGB-D observations, Hydra provides a spatial backbone for building persistent environment-level representations.

Hydra represents the environment as a layered 3D scene graph rather than only as a set of detected objects. The graph connects metric geometry, objects, agents, places, rooms, and building-level structure. This allows the system to maintain both low-level geometric information and higher-level semantic and topological abstractions, as illustrated in Figure 2.7.

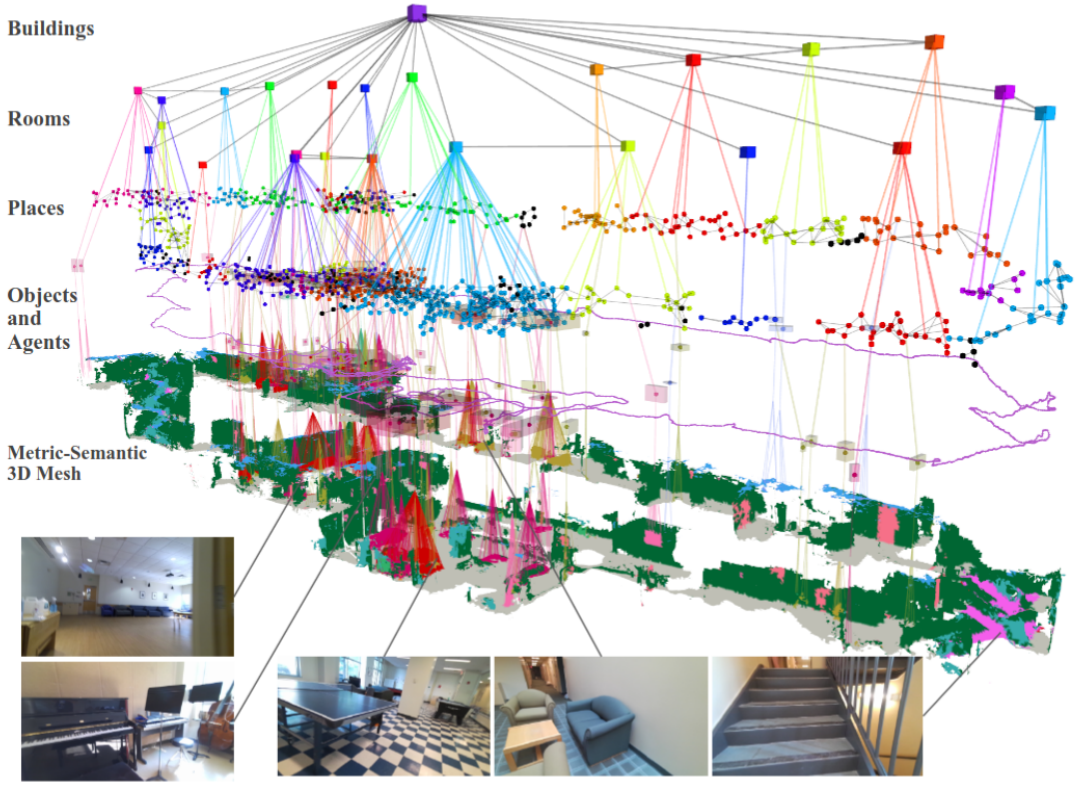


Figure 2.7.: Example of a layered 3D scene graph generated by Hydra, showing abstraction levels from metric-semantic 3D mesh to objects and agents, places, rooms, and buildings. Reproduced from [7].

2.5.1. Input Data and Processing Overview

Hydra operates on continuous sensor data from a moving robot. In the architecture described by Hughes et al., the raw inputs are images and IMU measurements, which are processed by low-level perception modules to estimate robot motion, reconstruct depth, and obtain semantic segmentation [7]. The motion estimate provides the transformations required to place observations into a common spatial reference frame, while depth information provides the 3D structure of the environment.

Semantic information is provided before the 3D scene graph layers are constructed. In simulation, these semantic labels can come from ground-truth annotations generated by the simulated environment. In real-world experiments, Hydra uses HRNet for semantic segmentation with labels from the ADE20K scene parsing dataset [7], [28], [29]. These semantic labels are combined with depth information to construct a metric-semantic 3D representation, where reconstructed geometry is associated with semantic classes such as wall, floor, furniture, or other object categories.

The overall processing pipeline transforms sensor-level data into graph-level spatial knowledge. Low-level perception estimates motion, depth, and semantic labels. Mid-level perception constructs the metric-semantic mesh, object layer, and places graph. High-level perception performs loop closure detection, scene graph optimization, and room detection. In this way, Hydra integrates observations over time to build a persistent layered representation of both the current surroundings and previously explored parts of the environment.

2.5.2. Layered 3D Scene Graph Representation

The main output of Hydra is a layered 3D scene graph, where nodes represent spatial concepts at different levels of abstraction and edges represent relationships between them. The lowest layer is the metric-semantic 3D mesh, which represents the physical geometry of the environment together with semantic labels. Above this, the object and agent layer represents detected objects and the robot trajectory. Object nodes can include semantic labels, centroids, and bounding boxes, while the agent is represented by a pose graph describing the robot's motion [7].

The places layer provides a sparse topological representation of obstacle-free space. A place represents a traversable location in the environment, and an edge between two places indicates straight-line traversability. The room layer groups places into larger spatial regions, while the building layer provides the highest-level abstraction. This hierarchy is useful because different robotic tasks require different levels of detail: metric geometry supports local navigation, places support topological reasoning, and objects and rooms support semantic interpretation.

Hydra separates scene graph construction into a frontend and a backend. The frontend incrementally assembles an odometric scene graph from the latest mesh, object, places, and agent-pose information. The backend corrects this representation when loop closures are detected by optimizing selected graph elements and propagating the correction to the remaining layers [7].

This separation allows continuous local graph construction while still supporting slower global correction.

2.5.3. Incremental Construction of Spatial Layers

Hydra constructs the scene graph incrementally rather than rebuilding the full environment in a batch process. A key design choice is the use of a local active window around the robot. Within this active window, Hydra maintains volumetric spatial representations such as a Truncated Signed Distance Field (TSDF) and an Euclidean Signed Distance Function (ESDF). This limits memory usage because the system does not need to maintain an ESDF of the entire environment at all times [7].

The metric-semantic mesh is extracted from the local volumetric representation. Object nodes are then generated by clustering mesh vertices that belong to the same semantic class. For each putative object, Hydra retains the semantic label inherited from the metric-semantic mesh and estimates geometric properties such as the centroid and bounding box. New observations can be merged with existing object nodes if they overlap spatially and belong to the same semantic class; otherwise, they are inserted as new object nodes.

The places layer is extracted from the ESDF using a Generalized Voronoi Diagram (GVD). The ESDF stores the distance from each voxel, or small 3D map cell, to the nearest obstacle. Voxels that are approximately equidistant from multiple obstacles form the GVD, which lies near the medial axis of free space. In indoor environments, this means that GVD voxels tend to lie near the centre of corridors, rooms, and passages rather than close to walls or obstacles. Hydra sparsifies this structure into a graph of places by selecting representative GVD voxels as place nodes and connecting them with edges when straight-line traversal is possible. Therefore, places are not object detections; they are topological free-space locations extracted from the geometry of the environment [7].

Room detection is then performed using the places graph. The key idea is that rooms can be identified by analysing how free-space connectivity changes when obstacles are virtually inflated. When the obstacle dilation distance increases, narrow openings such as doors gradually close. This causes the places graph to split into connected components, which can correspond to individual rooms. Hydra evaluates multiple dilation distances, selects a stable partition of the places graph, and assigns remaining unlabeled places using graph clustering based on modularity. In this way, room-level structure is extracted from the topology of free space rather than from direct image-based room classification or from processing the complete dense map [7].

2.5.4. Loop Closure, Optimization, and Real-Time Architecture

As a robot moves through an environment, odometry errors can accumulate and cause drift in the estimated map and scene graph. Hydra addresses this through loop closure detection and

scene graph optimization. Loop closure detection uses hierarchical descriptors derived from multiple graph layers, including visual appearance, nearby object semantics, and places geometry. Candidate loop closures are then geometrically verified using visual features or object-based registration [7].

Once a loop closure is accepted, Hydra optimizes the scene graph using an embedded deformation graph. This deformation graph contains selected elements such as the agent pose graph, mesh control points, and places layer. The optimization corrects accumulated drift and propagates the correction to the rest of the scene graph. Object centroids and bounding boxes can be recomputed after deformation, overlapping nodes can be merged, and rooms can be updated from the corrected places graph.

Hydra is organized as a highly parallelized architecture with low-level, mid-level, and high-level perception processes. Low-level perception operates close to sensor rate and includes visual-inertial odometry, semantic segmentation, and depth reconstruction. Mid-level perception constructs local spatial structures such as the mesh, object layer, places graph, and scene graph frontend. High-level perception performs loop closure detection, backend optimization, and room detection. This separation prevents slower global operations from blocking faster local perception, which is important for near-real-time robotic operation.

2.5.5. Relevance to the Proposed Framework

Hydra provides a strong methodological reference for this thesis because it demonstrates how a robot can construct a persistent spatial scene graph in real time. The most relevant ideas are the layered graph representation, incremental construction of spatial layers, local active-window mapping, places extraction from free-space topology, room detection from the places graph, loop closure correction, and separation between fast local processing and slower global optimization.

In this thesis, Hydra is useful as a spatial backbone for extending the existing risk-annotated scene graph framework. The baseline framework provides semantic and risk-oriented interpretation, while Hydra provides spatial consistency, scalability, and loop closure support. Combining these ideas motivates a spatial risk-annotated scene graph that can represent risk-relevant objects, semantic relations, uncertainty, and spatial context across larger robotic environments.

2.6. Risk-Annotated Scene Graph Framework Baseline

The framework developed in this thesis extends the risk-annotated scene graph framework presented by Schiel [1], which is referred to as the baseline in the following. This section outlines its processing pipeline and external interface, explains how the scene graph is constructed and annotated with risk-relevant information, and identifies the limitations that motivate the present work. The focus lies on the conceptual design of the framework rather than on its implementation.

The underlying segmentation, retrieval, and vision-language techniques are introduced separately in Sections 2.7 and 2.9.

2.6.1. Information Flow in the Baseline Framework

The baseline framework follows a modular, worker-based processing pipeline in which RGB-D data is received, processed, interpreted, and integrated into a risk-annotated scene graph. The input to the system is an RGB-D observation of the environment, consisting of a colour image and its corresponding depth image, which is either provided through a dataset or submitted through the FastAPI-based interface described in Section 2.6.2. This design separates data input and client communication from the perception and scene graph generation modules. The resulting information flow is illustrated in Figure 2.8.

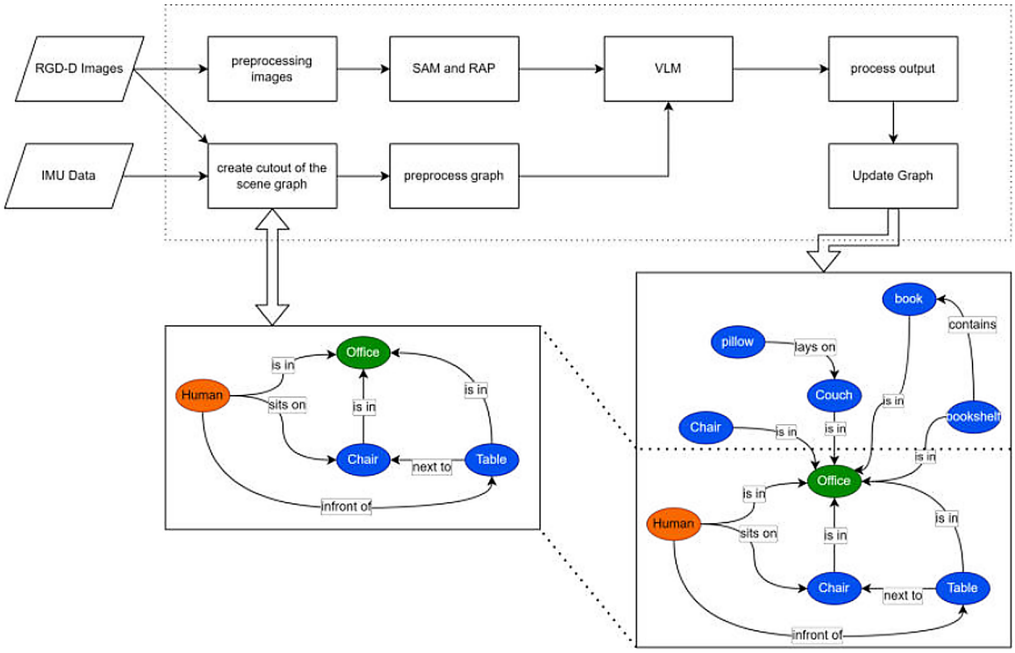


Figure 2.8.: Information flow of the existing risk-annotated scene graph framework. Adapted from [1].

Each received sample is handled as a task in the processing pipeline. The colour image is first segmented with the Segment Anything Model (SAM) [30] to obtain class-agnostic object candidates. For every retained candidate, the framework stores an image crop, a bounding box, a binary mask, and a segmentation quality score. These candidates carry no semantic meaning, so their identities must be determined in the subsequent stages. The depth image, camera parameters, and pose information are used for the later spatial processing, including coordinate conversion, visibility checking, and the spatial grounding of graph elements.

The object crops are then passed to the Retrieval-Augmented Perception (RAP) module, which returns label candidates from a visual knowledge base of previously stored observations (Section 2.9.1). In parallel, the existing scene graph is reduced to a cutout that contains only the nodes and edges likely to be visible in the current camera view, determined from the current pose and depth image. The RGB image, the scene-graph cutout, and the RAP label candidates are subsequently passed to a vision-language model (VLM), which returns an updated graph for the current view (Section 2.9.2). Finally, this output is grounded in 3D and merged into the complete scene graph, as described in Section 2.6.3.

2.6.2. FastAPI

FastAPI is a Python web framework for building application programming interfaces (APIs), through which external clients communicate with a software system via network-accessible endpoints, typically using HTTP requests [31]. The expected format of incoming requests can be defined explicitly, which suits complex inputs such as RGB-D images, timestamps, and pose information, and asynchronous operation allows several requests to be handled concurrently [31]. In the baseline framework, FastAPI exposes the scene graph generation pipeline as an external service and keeps client applications separated from the internal perception, segmentation, VLM, and scene graph modules. For this external communication, the internal graph is converted into a dictionary-based representation in which node and edge information are stored separately.

This interface is well suited to dataset-based testing and to request-response communication with external clients [31]. Robotic software, however, relies on continuous high-frequency sensor streams, synchronised multi-modal data, distributed nodes, and predictable timing behaviour, for which a REST-based interface is less natural. FastAPI is therefore not ideal as the primary communication mechanism for deployment on a robotic platform, which motivates the port of the framework to ROS 2 [10].

2.6.3. Scene Graph Construction and Risk Annotation

The baseline framework represents the scene graph as a directed graph, where entities in the environment are modelled as nodes and relationships between entities are modelled as edges [1]. The implementation uses NetworkX, a Python library for network analysis and graph manipulation, which supports incremental graph operations such as adding, updating, and removing nodes and edges [32].

Each node stores semantic, spatial, and risk-related information. Important node attributes include a persistent identifier, object label, semantic description, semantic pose, 3D position, risk metadata, appearance score, confidence score, and layer type. The semantic description describes the visual identity of the entity, while the semantic pose describes its current physical state, such as standing, lying, open, closed, or moving. Edges store relationships between nodes, including the source node, target node, relationship label, distance between the connected entities, and an appearance score.

After each VLM inference step, the returned graph is converted from image-space information into 3D coordinates using the depth image and camera pose. The converted output is then compared with the scene-graph cutout that was originally sent to the VLM, and the two are combined following an update-or-insert strategy. For each node in the VLM output, the framework checks whether a node with the same identifier already exists in the complete scene graph. If it exists, its attributes are updated using the latest VLM output and its appearance score is increased. Otherwise, it is inserted as a new node with an initial appearance score. This allows newly detected objects or humans to be added while persistent entities are reinforced over time.

Nodes that were present in the VLM input cutout but are missing from the VLM output are interpreted as not being detected in the current update. Instead of being removed immediately, their appearance score is decreased, and a node is removed only when its appearance score falls below a predefined threshold. This prevents isolated missed detections from immediately deleting objects from the graph.

The same principle is applied to edges. If an edge returned by the VLM already exists between the corresponding source and target nodes, its relationship label and distance are updated and its appearance score is increased; otherwise, a new edge is added with an initial appearance score. The edge distance is calculated from the 3D positions of the connected nodes, so that spatial relationships are stored together with semantic relationship labels. Edges that were present in the input cutout but are missing from the VLM output are weakened by decreasing their appearance score and are removed once it falls below the threshold. Temporal consistency is therefore handled through appearance-based persistence: repeatedly observed nodes and edges become more stable, while missing or transient entities are gradually removed.

2.6.4. Limitations of the Baseline Framework

When considered for deployment on a mobile robot, the baseline framework has limitations in three areas: its integration into the robotic system and the spatial extent it can represent, the reliability of the information stored in the scene graph, and its computational cost.

The first limitation arises from the service-oriented interface of the framework, which is not aligned with robot-native communication patterns (Section 2.6.2) and therefore makes deployment on real robotic platforms less straightforward. The spatial extent of the representation is likewise restricted, as the scene graph is updated based only on the current frame, camera pose, and visible graph elements. This is suitable for local scene understanding, but it provides no mechanism for keeping the graph spatially consistent across multiple rooms, corridors, or revisited locations. Since the framework also lacks loop closure, an area revisited after accumulated pose drift may lead to revisited objects being inserted as new observations, or to updates based on spatial information that is no longer globally consistent. Anchoring the scene graph in a SLAM-based spatial map with loop closure, such as the one provided by Hydra [7], is therefore necessary to extend the framework beyond local frame-by-frame scene understanding.

Beyond these integration aspects, the information stored in the scene graph is only partially reliable. Uncertainty is expressed solely through confidence values and appearance scores. Appearance-based persistence distinguishes repeatedly observed entities from briefly observed ones, but it maintains neither a probability distribution over object classes or relationship labels nor a probabilistic estimate of object existence or spatial position, although such information is required for downstream risk assessment (Section 2.8.4). The relationships stored on the edges are similarly limited. They are inferred by the VLM from a single two-dimensional view, and several of them, such as *in front of* or *behind*, are defined relative to the observer rather than to the scene. An object that is behind another object from one viewpoint may appear beside or in front of it from another, so the same pair of objects can receive contradictory labels as the robot moves, and each update overwrites the previously stored label. Such egocentric relations do not describe a stable property of the three-dimensional scene. Moreover, the scene graph depends directly on the VLM output and therefore inherits the failure modes described in Section 2.9.2. Since the parsing and graph-combination steps mitigate only some of these effects, the VLM output must be bounded, checked, and converted into a consistent graph representation before it can serve as reliable input for risk assessment.

Finally, the runtime performance of the framework is constrained by its per-frame inference load. Segmenting every frame with SAM in automatic mode and invoking the VLM on every frame limits the achievable update rate of the scene graph on embedded hardware, for the reasons given in Sections 2.7.1 and 2.9.2. Together, these limitations motivate the extensions proposed in this thesis: porting the framework to ROS 2, anchoring the scene graph in a SLAM-based spatial map with loop closure, introducing a probabilistic object representation, deriving object relationships from the three-dimensional geometry of the scene, constraining and validating the VLM output, and reducing the per-frame inference load to enable near-real-time operation.

2.7. Lightweight Promptable Segmentation with NanoSAM

The baseline framework uses the Segment Anything Model (SAM) to generate class-agnostic object candidates from the RGB component of each observation [1], [30]. Although SAM achieves strong zero-shot segmentation quality, its image encoder is computationally expensive. On an embedded robotic platform, segmentation must share the available compute and memory with spatial mapping, retrieval, and vision-language model inference, which makes the original model unsuitable for near-real-time operation. In this thesis, SAM is therefore replaced by NanoSAM, a distilled SAM variant developed by NVIDIA for real-time inference on Jetson devices using NVIDIA TensorRT [33], [34]. This section introduces the architecture of SAM that NanoSAM inherits, the knowledge distillation approach used to obtain NanoSAM, its deployment and performance characteristics, and its use in the proposed framework.

2.7.1. Architecture of the Segment Anything Model

SAM formulates segmentation as a promptable task: given an image and a prompt that indicates what to segment, the model returns a valid segmentation mask, even if the prompt is ambiguous [30]. The model is class-agnostic, meaning that it delineates image regions without assigning semantic labels. It consists of three components. The image encoder is a Vision Transformer (ViT) [35], pre-trained as a masked autoencoder [36], which processes an input image resized and padded to 1024×1024 pixels. The image is divided into 16×16 pixel patches, which are processed by transformer blocks using mostly windowed and occasionally global self-attention, and a convolutional neck compresses the result into a dense image embedding of $64 \times 64 \times 256$.

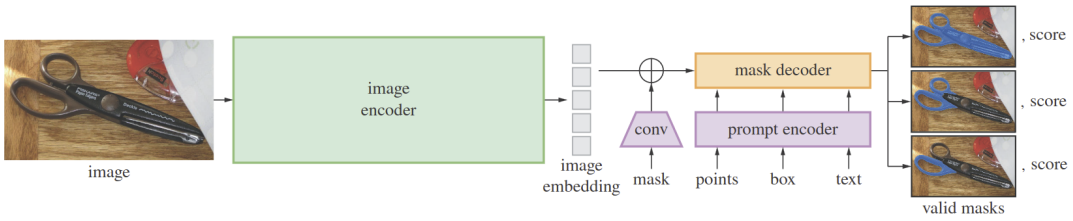


Figure 2.9.: Overview of the Segment Anything Model (SAM) architecture. The model consists of an image encoder, a prompt encoder, and a lightweight mask decoder that produces segmentation masks conditioned on input prompts. Reproduced from Kirillov et al. [30].

In this work, the model is prompted exclusively with single foreground points placed on a regular grid. The prompt encoder maps each point to a 256-dimensional token by combining a random Fourier positional encoding of its image coordinates with a learned embedding identifying it as a foreground point. Since the same positional encoding is applied to the locations of the image embedding, the prompt token and the image features share a common coordinate representation, which allows the decoder to relate the prompt to the corresponding image region. The mask decoder is a lightweight two-way transformer, in which the prompt token and a set of learned output tokens attend to each other and to the image embedding, while the image embedding in turn attends to the tokens. After upsampling the updated embedding, each output mask token is mapped to a per-pixel classifier that yields mask logits, and a dedicated output token predicts the intersection-over-union (IoU) of each mask with the target object [30].

Two design properties of SAM are particularly relevant for this thesis. First, the computational cost is distributed asymmetrically: the image encoder accounts for the large majority of the computation but is executed only once per image, after which the embedding can be reused for an arbitrary number of prompts at low additional cost. Second, since a single point can be ambiguous, for example when it lies on a part of an object as well as on the object as a whole, the decoder predicts several candidate masks at different levels of granularity together with their predicted IoU. For automatic segmentation without user input, SAM prompts the image with a regular grid of foreground points and filters the resulting masks by predicted IoU, mask stability, and non-maximum suppression (NMS). This automatic mode multiplies the decoder cost by

the number of grid points and, combined with the large ViT encoder, is not compatible with near-real-time operation on embedded hardware.

2.7.2. Knowledge Distillation and MobileSAM

Knowledge distillation is a model compression technique in which a compact student network is trained to reproduce the outputs or intermediate representations of a larger teacher network [37]. Since the student is supervised by the teacher rather than by ground-truth annotations, a substantial part of the teacher's behaviour can be transferred to a model with considerably lower computational cost.

MobileSAM applies this principle to SAM [38]. Zhang et al. observed that jointly distilling the image encoder and the mask decoder is difficult, because both components are optimised in a coupled manner. They therefore proposed decoupled distillation, in which only the image encoder is distilled: a lightweight TinyViT-based encoder [39] is trained to regress the image embeddings of SAM's ViT-H encoder. Because the student produces embeddings in the same format as the teacher, the original prompt encoder and mask decoder can be reused without retraining. In this way, MobileSAM reduces the image encoder from more than 600 million to approximately five million parameters while retaining segmentation behaviour comparable to the original model [38].

2.7.3. NanoSAM

NanoSAM extends the decoupled distillation strategy of MobileSAM by a further compression step targeted at embedded NVIDIA hardware [33]. The MobileSAM image encoder acts as the teacher, and a convolutional network acts as the student. The student uses an ImageNet-initialised ResNet-18 backbone [40], which processes the 1024×1024 input through a stem and four stages of residual blocks, progressively halving the spatial resolution and doubling the number of channels until a feature map of $32 \times 32 \times 512$ is obtained. A convolutional neck then reduces the channels, upsamples the features to 64×64 using a transposed convolution, and projects them to 256 channels. Finally, a learned positional embedding of $64 \times 64 \times 256$ is added. This positional embedding compensates for the largely translation-equivariant nature of convolutions, which would otherwise prevent the student from reproducing the position-dependent component of the transformer teacher's embedding.

The student is trained on unlabelled images from the COCO 2017 training set [41] by regressing the teacher embedding with a Huber loss [42]. Since no segmentation annotations are required, the distillation relies entirely on the teacher's representations. Because the embedding format is preserved, NanoSAM reuses the MobileSAM prompt encoder and mask decoder without modification. The resulting pipeline therefore consists of the distilled ResNet-18 encoder followed by the unchanged MobileSAM decoder.

For deployment, both components are compiled into TensorRT engines. TensorRT converts a trained network into a hardware-specific engine by applying, among other techniques, layer fusion, reduced-precision execution, and automatic selection of kernel implementations for the target GPU [34]. The NanoSAM encoder is executed in half precision (FP16), whereas the decoder is executed in single precision (FP32) [33]. Since the engines are specific to the GPU architecture and TensorRT version, they must be built on, or for, the target platform. Table 2.1 summarises the latency and accuracy reported by the developers, where accuracy is measured by prompting the model with ground-truth bounding boxes from the COCO 2017 validation set.

Table 2.1.: Latency on Jetson AGX Orin and mean IoU on COCO 2017 validation with ground-truth box prompts, as reported in [33].

	Latency (ms)		mIoU		
Model	Encoder	Full pipeline	All	Small	Large
MobileSAM	35.0	39.0	0.728	0.658	0.804
NanoSAM (ResNet-18)	4.2	8.1	0.706	0.624	0.796

2.7.4. Limitations

NanoSAM has two limitations that affect its use in the proposed framework. First, although NanoSAM uses the same decoder architecture as SAM, its TensorRT-based predictor only exposes prompt-based inference and does not include the automatic mask generation pipeline of SAM, which therefore has to be implemented separately as a post-processing wrapper around the model. In particular, object candidates must be obtained through externally generated prompts, and duplicate masks arising from neighbouring prompts must be removed by the application. Second, the distillation reduces segmentation accuracy, particularly for small objects, which may nevertheless be safety-relevant in cluttered indoor environments. Consequently, NanoSAM masks must be regarded as uncertain object proposals that require subsequent filtering, tracking, and semantic interpretation.

2.8. Multi-Object Tracking and Data Association

Segmentation, as discussed in Section 2.7, describes the objects visible in a single image, but it does not establish whether a segmented region corresponds to an object that has already been observed. For a mobile robot operating over an extended period, this distinction is essential. As the robot moves, the same physical object reappears from different viewpoints, at different distances, and

under varying degrees of occlusion. If every observation were treated independently, the scene representation would accumulate duplicate objects, and information gathered over time could not be attributed to the same physical entity. Multi-object tracking addresses this problem by maintaining persistent object representations over time and relating each new observation to them.

2.8.1. The Data Association Problem

The central task in multi-object tracking is data association: deciding which current observation corresponds to which previously established object. If an observation is sufficiently consistent with an existing object representation, it is associated with that object and contributes to its refinement. Otherwise, it is treated as a previously unseen instance and used to initialise a new object representation. Conversely, an existing object that is not observed in the current frame is not necessarily absent, since it may be occluded, outside the field of view, or missed by the perception system.

The reliability of this decision depends on the evidence used to compare observations with existing objects. Image-space evidence, such as the position or overlap of regions in the image, changes substantially when the camera moves, even if the observed object remains stationary. A table, for example, can occupy entirely different image regions when viewed from opposite sides of a room. For a mobile robot, apparent motion in the image is therefore caused by both object motion and ego-motion. When depth measurements and camera poses are available, observations can instead be expressed in a common three-dimensional reference frame, in which the position and extent of a static object remain approximately constant across viewpoints. Three-dimensional information therefore provides more stable evidence for associating observations that are far apart in time or taken from very different viewpoints.

Two types of association error must be considered. A false association merges observations of distinct physical objects into a single representation, while a missed association splits one physical object into duplicate representations. Both errors propagate directly into any scene representation built on top of the tracking result.

2.8.2. Object-Level SLAM with Fusion++

A relevant precedent for persistent, object-centric scene representation is Fusion++, an object-level SLAM system proposed by McCormac et al. [43]. Instead of representing the environment exclusively through a single monolithic geometric model, Fusion++ builds a persistent map composed of individual object instances. The system operates on RGB-D input and uses Mask R-CNN instance segmentation [44] to detect object regions in incoming frames. The corresponding depth measurements are used to initialise a separate volumetric TSDF model for each newly

detected object, with the voxel resolution of each model chosen according to the object's size. A coarse TSDF of the background is maintained in parallel to support camera tracking [43].

Data association in Fusion++ couples the persistent 3D models with the 2D segmentation output. Existing object models are rendered into the current camera view, and each new instance mask is compared with these renderings using their intersection-over-union. A mask is associated with an existing object if the overlap exceeds a threshold. Otherwise, it initialises a new object [43]. When an object is observed again, the new depth measurements are fused into its existing TSDF rather than creating an independent copy. Since an object is rarely fully visible from a single viewpoint, repeated observations from different viewpoints gradually complete its geometry. Persistent association therefore serves two purposes: it preserves object identity, and it allows the spatial representation of each object to be refined over time.

Fusion++ also maintains uncertainty-related quantities for each object. Semantic identity is not fixed at initialisation. Instead, the class probabilities predicted by Mask R-CNN are fused over successive observations, so that the class estimate of an object can change as further evidence becomes available, independently of the object's identity in the map. In addition, each object carries an existence probability, which is derived from how often the object is detected compared with how often it is expected to be visible but is not detected. Objects whose existence probability falls too low are removed, which suppresses spurious detections without discarding objects after a single missed observation [43]. Finally, the object instances are organised in a six-degree-of-freedom pose graph and used as landmarks for tracking, relocalisation, and loop-closure detection. As a result, objects become persistent elements of the spatial map rather than transient detections tied to individual frames.

2.8.3. Limitations of Fusion++

Despite these properties, Fusion++ is not suitable as the basis of the proposed framework. Its semantic perception is closed-set, since objects are detected and classified by Mask R-CNN, which is trained on a fixed set of categories [43]. Objects outside this set are not represented as distinct entities, which conflicts with the requirement of open-vocabulary identification of risk-relevant objects. Furthermore, Fusion++ produces an object-level map rather than a scene graph. It contains no relationships between objects, descriptive object properties, or higher-level spatial structures such as rooms, and therefore lacks the relational context on which downstream safety reasoning depends.

In addition, Fusion++ assumes a static environment [43]. Its existence probability indicates whether an object is repeatedly detected, but not whether a previously observed object is still at its last observed position, even though this uncertainty is itself risk-relevant in dynamic environments. Finally, maintaining a separate volumetric model for every object, combined with a two-stage segmentation network, results in memory and computational costs that grow with the

number of objects. This conflicts with the requirement of near-real-time operation on embedded hardware.

2.8.4. Relevance for Persistent Scene Representations

Despite these limitations, the central idea behind Fusion++ remains relevant to this work. Image observations can be combined with depth to build persistent object representations, and new observations can be linked back to them over time. For a scene graph that supports risk assessment, this means that each node should represent a physical object rather than a single detection. Information collected at different times, such as labels, properties, or positions, can then be gathered in one place instead of being spread across duplicate entries. The semantic label of an object should also be treated as an estimate that can improve with more observations, rather than as a fixed value assigned when the object is first seen.

In the same way, the presence of an object is better described by a probability than by a simple yes or no. This makes it possible to tell apart objects that have been seen many times, objects that were seen only briefly, and objects that have not been observed for some time. Such distinctions matter for risk assessment, since an object that may have moved should be treated differently from one whose position is well known. These ideas, however, need to be applied under conditions that Fusion++ does not consider, including open-vocabulary object identification, relations between objects, changing object positions, and near-real-time operation on embedded hardware.

2.9. Visual Retrieval and Vision-Language Models

Class-agnostic segmentation produces object masks without semantic meaning. Labels can be assigned to such masks by visual retrieval, which reuses the label of a previously seen, similar object, or by a vision-language model (VLM), which names an object directly from its appearance. Retrieval is computationally cheap but limited to what has been observed before, whereas a VLM can describe arbitrary objects at a considerably higher computational cost. The baseline framework combines both approaches [1], and its concept forms the starting point for the pipeline developed in this thesis.

2.9.1. Retrieval-Augmented Perception

Visual retrieval requires an image encoder that converts an image into a fixed-length vector, referred to as an embedding, such that similar images obtain similar embeddings. A widely used encoder is the image encoder of Contrastive Language–Image Pre-training (CLIP) [45]. In its ViT-B/32 variant, the image is resized to 224×224 pixels, divided into 32×32 pixel patches, and

processed by a Vision Transformer [35]. A summary token that collects information from all patches is projected to a vector of 512 values and normalised to unit length.

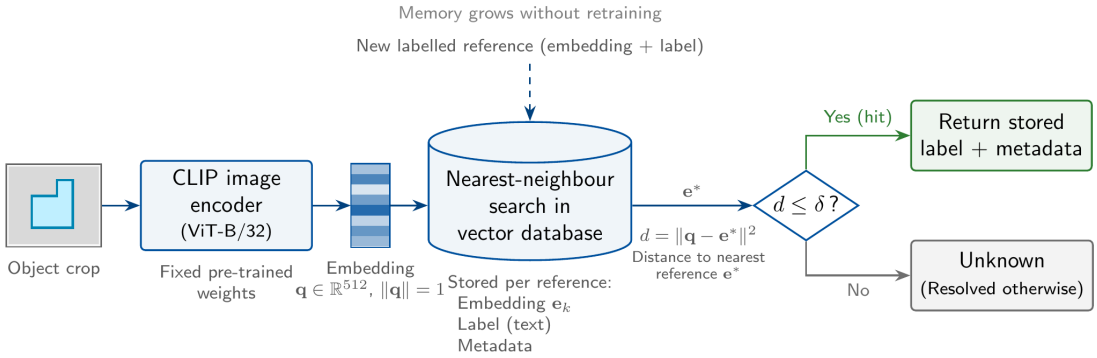


Figure 2.10.: Principle of visual retrieval: the crop embedding is compared with stored references, and the nearest label is returned if its distance lies below the threshold δ.

The meaning of these embeddings results from the encoder weights, which were determined once during pre-training and remain fixed afterwards. CLIP was trained on approximately 400 million images with their captions; during training only, a separate text encoder maps each caption into the same space, and both networks are optimised so that each image lies close to its own caption and far from all others [45]. To achieve this, the image encoder must extract features that describe objects, materials, and scenes. Images with similar content must lie close to similar captions and are therefore also close to each other. After training, images are compared using the image encoder alone. To compare a new image with a stored reference image, their embeddings are compared. Let $\mathbf{q}$ denote the embedding of the new image (the query) and $\mathbf{e}$ the embedding of a stored reference. Their squared Euclidean distance $\|\mathbf{q} - \mathbf{e}\|^2$, i.e. the sum of the squared differences of all 512 values, is directly related to their cosine similarity $\mathbf{q}^\top \mathbf{e}$. Since both embeddings have unit length,

$$\|\mathbf{q} - \mathbf{e}\|^2 = \|\mathbf{q}\|^2 + \|\mathbf{e}\|^2 - 2\,\mathbf{q}^\top \mathbf{e} = 2\left(1 - \mathbf{q}^\top \mathbf{e}\right). \tag{2.1}$$

The distance is therefore zero for identical embeddings and increases as the cosine similarity decreases, so both measures rank stored references in the same order.

Retrieval-augmented methods use stored knowledge to support the interpretation of new input [46]. In visual retrieval, a memory stores the embedding of each labelled reference image together with its label as plain text and optional metadata. Vector databases such as ChromaDB organise these embeddings in an approximate nearest-neighbour index for efficient search [47]. A new image is embedded with the same encoder, and if the distance to the nearest reference lies below a threshold, the stored label is returned; otherwise, the image remains unknown. The memory thus does not interpret images itself but determines whether an image closely resembles one that has already been labelled. New objects can be added at any time without retraining, since the learning takes place in the stored references rather than in the encoder.

In the baseline framework, this principle is implemented as Retrieval-Augmented Perception (RAP) [1]. For each frame, the object cutouts generated by the Segment Anything Model are embedded with CLIP and compared with a visual knowledge base in ChromaDB. The labels of sufficiently similar references are passed to the VLM as label candidates, which guide its interpretation and help to keep labels consistent across observations. RAP therefore supports the VLM in every inference step but does not replace it. The same principle could also be used to resolve an object directly when a sufficiently similar reference exists, so that the VLM would only be required for the remaining objects.

Retrieval is fast, and repeated observations of the same object receive consistent labels. However, it can only return labels that are already stored, an incorrect entry is propagated to all similar observations, and the threshold trades the number of resolved objects against the number of accepted errors. Moreover, the encoder summarises the entire image including its background, because nothing indicates which pixels belong to the object. Large uniform surfaces such as walls and ceilings in the same room can therefore appear more similar to each other than to the same surface type elsewhere.

2.9.2. Vision-Language Models

A VLM consists of a vision encoder, which converts image patches into feature vectors, a connector, which maps these features into the input space of a large language model, and the language model, which generates the answer token by token from the visual input and a text prompt [48]. Since each token requires one forward pass, the latency grows with the length of the answer. A pre-trained VLM can be applied to new tasks without retraining, as the task is specified in the prompt.

Compared with conventional object detectors and segmentation networks, which are trained on a fixed set of classes, VLMs are open-vocabulary: they can name objects, materials, and states that were never defined as classes in advance, and they can describe properties beyond the object category, such as the condition of an object or its relation to its surroundings. This capability results from pre-training on large and diverse image–text data. At the same time, the generated text is not restricted to what is visible in the image. A well-documented failure mode is object hallucination, in which a model describes objects or attributes that are not present, often because they frequently co-occur with the visible content in the training data [49]. The flexibility of VLMs is therefore accompanied by the risk of plausible but incorrect answers.

In the baseline framework, the VLM is the central reasoning component [1]. For each frame, it receives the RGB image, the part of the existing scene graph expected to be visible, and the label candidates from RAP, and returns an updated scene graph with labels, descriptions, states, confidence values, risk-relevant properties, and relationships. The output is parsed, converted into 3D coordinates, and merged into the scene graph. The framework is model-interchangeable, and the study reports formatting errors, repetitions, hallucinated relationships, and unstable location output as typical difficulties [1]. Since the VLM is called for every frame, its inference time also limits the update rate of the scene graph.

In general, VLM output cannot be treated as ground truth. A reported confidence value is generated text rather than a calibrated probability, and models often state high confidence for incorrect answers [50]. VLM output should therefore be validated, for example against confidence thresholds, before it is stored in a persistent environment representation.

2.9.3. The Qwen Vision-Language Model Family

The baseline study evaluated three VLMs for scene graph generation: LLaVA 1.6 Mistral 7B, Qwen2.5-VL 7B, and Phi-4 Multimodal [1]. The models were compared with respect to the compatibility of their input and output formats, their tendency towards repetition and hallucination, and the stability of the generated object locations. Based on this comparison, the Qwen series was identified as the most suitable model family, and later tests of the baseline framework used the newer Qwen3-VL generation, taking advantage of the model-interchangeable design of the framework [1]. This thesis builds on this result. Instead of repeating the comparison between model families, it continues with the best-performing Qwen family and updates the model choice to the latest available generations, with a focus on small models that are suited to embedded hardware.

The Qwen models of Alibaba Cloud are published with open weights in several sizes and can therefore be executed locally on embedded hardware. Table 2.2 lists the relevant generations.

Table 2.2.: Release history of the open-weight Qwen vision-language models.

Generation	Release	Small dense sizes
Qwen2-VL	Aug. 2024	2B, 7B
Qwen2.5-VL	Jan. 2025	3B, 7B
Qwen3-VL	Oct. 2025	2B, 4B, 8B
Qwen3.5	Mar. 2026	0.8B, 2B, 4B, 9B

Qwen2-VL introduced image processing at native resolution [51], and Qwen2.5-VL improved efficiency, object localisation, and document understanding [52]. Qwen3-VL builds on the Qwen3 language models and supports contexts of up to 256K tokens [53]. Qwen3.5 is trained natively on mixed image and text data instead of attaching a vision encoder to a separately trained language model, and by default generates a reasoning trace before its answer, which can improve accuracy but increases latency [54].

The released models differ in size, architecture, and operating mode. Dense models use all of their parameters for every token, whereas mixture-of-experts models activate only a subset of their parameters per token, which reduces computation but not the memory required to store the full model [53]. In addition, Qwen3-VL is released in an instruction-following variant, which answers directly, and a thinking variant, which reasons explicitly before answering [53], while Qwen3.5 combines both behaviours in one model [54]. For embedded deployment, the memory footprint is decisive. The memory required for the weights is approximately the number of parameters

multiplied by the number of bits per parameter, so that an 8B model in 16-bit precision requires about 16 GB, whereas a 4B model stored with about 4 bits per parameter requires roughly 2 to 3 GB. Small dense models, which are commonly also distributed in such reduced-precision formats, are therefore the most suitable candidates when the VLM has to run alongside other perception and mapping components on the same device.

Since the baseline study, a new generation has appeared roughly every six to twelve months, and small models have improved noticeably: on the MMMU-Pro benchmark, Qwen3.5-4B scores 65.4 % compared with 52.0 % for Qwen3-VL-4B [55]. Although general benchmarks do not directly predict task-specific accuracy, this trend indicates that the small Qwen3-VL and Qwen3.5 models are promising candidates for the present work, and that future generations may improve the results further. The model-interchangeable design of the baseline framework allows such newer models to be adopted without changes to the remaining pipeline.

2.10. Embedded Computing Hardware: NVIDIA Jetson Platforms

The proposed framework combines several computationally demanding components, including NanoSAM-based segmentation, retrieval-augmented perception, VLM reasoning, Hydra-based spatial mapping, and risk-annotated scene graph generation. For onboard robotic deployment, these components must run on embedded hardware with sufficient AI compute, memory capacity, and energy efficiency. This section introduces the two NVIDIA Jetson platforms relevant to this work, the Jetson AGX Orin Developer Kit and the Jetson AGX Thor Developer Kit.

2.10.1. NVIDIA Jetson AGX Orin Developer Kit

NVIDIA Jetson AGX Orin is an embedded edge-AI computing platform designed for robotics, computer vision, and autonomous machines. Like other members of the Jetson family, it integrates CPU, GPU, memory, I/O, and AI acceleration into a compact module for onboard computation. Jetson AGX Orin is based on the NVIDIA Ampere architecture, combining a 2048-core GPU with 64 Tensor Cores and a 12-core Arm Cortex-A78AE CPU. It provides up to 275 TOPS of AI performance, 64 GB of memory, and a configurable power range of 15 W to 60 W [56].

Jetson AGX Orin was used as the target platform in the baseline framework [57]. Its compute capacity is sufficient for executing individual perception and reasoning components, such as segmentation and VLM inference with models in the 7B parameter class. However, executing the complete pipeline, consisting of segmentation, retrieval, VLM reasoning, and spatial mapping, concurrently on a single module can be expected to place considerable demands on the available memory and compute resources. Since the CPU and GPU share the same physical memory and power budget, these components compete for the same resources, which may limit the achievable update rate of the scene graph.

2.10.2. NVIDIA Jetson AGX Thor Developer Kit

NVIDIA Jetson AGX Thor is the successor platform within the Jetson family and is designed specifically for physical AI, robotics, and real-time multimodal perception. It is based on the NVIDIA Blackwell architecture, combining a 2560-core GPU with 96 fifth-generation Tensor Cores and a 14-core Arm Neoverse-V3AE CPU. It provides up to 2070 FP4 sparse TFLOPS of AI compute, 128 GB of memory, and a configurable power range of 40 W to 130 W [58]. According to the manufacturer, Jetson AGX Thor provides up to 7.5 times higher AI compute and 3.5 times better energy efficiency than Jetson AGX Orin [58].[1]

This additional capacity is potentially relevant for the proposed framework, as it may allow the perception, reasoning, and mapping components to be executed concurrently on the same embedded platform with fewer resource constraints. Jetson AGX Thor also offers improved support for transformer-based and multimodal workloads. Jetson AI Lab lists Qwen3-VL models, including Qwen3-VL 4B and Qwen3-VL 8B, as available for Jetson AGX Thor through Quick Start Runner and vLLM-based serving configurations [59]. Since the Qwen model series was identified as the best-performing option in the baseline framework [57], Jetson AGX Thor represents a suitable hardware basis for extending the system toward onboard robotic deployment without dependence on an external workstation for model inference.

2.11. Scene Graphs for Downstream Safety Reasoning

This section discusses how a generated scene graph can be used as structured input for downstream risk assessment. SafetyDetect by Mullen et al. is considered as an example of scene-graph-based safety reasoning [9]. The work proposes a dataset and LLM-based approach for detecting anomalous situations in home environments, such as food left outside storage, a stove left on, or harmful objects being accessible to children. In the proposed pipeline, object relations from a scene graph are provided to an LLM, which classifies situations as normal, dangerous, unsanitary, or dangerous for children.

The relevance of SafetyDetect lies in its demonstration that object relations can be important for safety-related reasoning. Many risks cannot be inferred from object identity alone, because the same object may be safe or unsafe depending on its relation to other objects, humans, or locations. A scene graph is therefore useful because it represents these relations explicitly and provides a structured intermediate representation for a downstream reasoning module.

At the same time, SafetyDetect highlights the importance of the scene graph generation stage. The authors state that their method is limited by perception, particularly by how the scene graph is created [9]. For example, a scene graph may represent that an object is inside a room and on the floor, but it may not indicate whether the object is blocking a doorway and creating a

[1] The compute figures of the two platforms are specified at different numerical precisions (FP4 sparse TFLOPS for Jetson AGX Thor and TOPS for Jetson AGX Orin). The comparison therefore reflects the manufacturer's specification rather than a measured result.

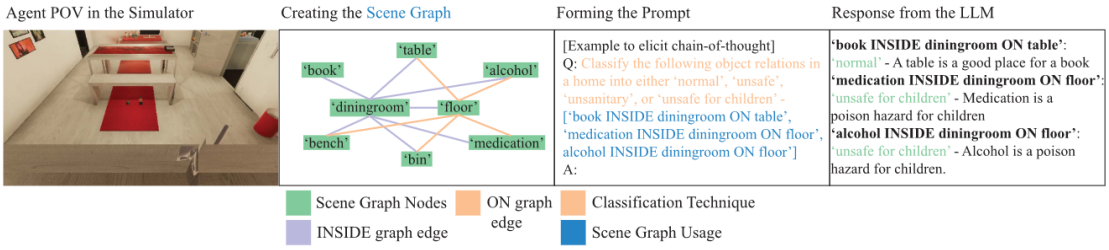


Figure 2.11.: SafetyDetect pipeline showing the agent point of view, scene graph construction, prompt formation, and LLM-based anomaly classification. Reproduced from [9].

tripping hazard. This shows that the downstream safety assessment depends not only on the reasoning capability of the LLM, but also on whether the scene graph contains the required objects, relations, spatial details, and obstruction information.

This limitation is important for the present thesis. SafetyDetect mainly focuses on reasoning after a scene graph has already been generated. In simulation, the scene graph is provided by the VirtualHome environment, while in the real-world TurtleBot experiment an existing scene graph generation method is used before the graph is passed to the LLM. Therefore, SafetyDetect is more relevant as an example of downstream safety reasoning than as a method for constructing a persistent spatial scene graph.

Consequently, SafetyDetect can be understood as evidence for the downstream value of scene-graph-based representations in safety reasoning. However, it also indicates that the effectiveness of such reasoning depends strongly on the quality of the generated scene graph. This supports the central motivation of the present thesis: improving scene graph generation, spatial grounding, persistence, and uncertainty representation is a necessary step toward more reliable dynamic risk assessment in robotic systems.

3. Methodology

This chapter describes the concept of the proposed framework and the reasoning behind its design. It explains what each component does and why it was designed in the chosen way, whereas software details, model serving, and parameter values are deferred to Chapter 4. Section 3.1 presents the overall architecture and its central design decisions. The subsequent sections follow the flow of data through the framework: the preparation of synchronised input frames (Section 3.2), segmentation and persistent object tracking (Section 3.3), semantic labelling (Section 3.4), the fusion of semantic information with the spatial scene graph (Section 3.5), the representation of uncertainty (Section 3.5.2), and the handling of loop closures and revisited environments (Section 3.6).

3.1. Framework Overview

This section presents the overall structure of the proposed framework before its components are described individually in the remainder of the chapter. The structure is derived from the four aspects defined in Section **??**, **??**, namely portability, spatial scalability, uncertainty awareness, and near-real-time operation, and from the limitations of the baseline framework [1] identified in Section 2.6.4, *Limitations of the Baseline Framework*. In essence, the framework combines Hydra as a spatial backbone with a semantic pipeline that identifies objects and characterises their mobility, and joins the two through persistent object identities. Section 3.1.1, *Framework Architecture*, introduces the components and follows the flow of data between them, and Section 3.1.2, *Central Design Decisions*, explains the design decisions behind this structure and relates each of them to the aspects and limitations it addresses.

3.1.1. Framework Architecture

The framework is realised as a set of ROS 2 nodes [10] that exchange data through the publish-subscribe mechanism introduced in Section 2.3. It consists of three nodes developed in this thesis, the preprocessor, the Phase 1 semantic coordinator, and the scene graph fuser, and of three external software components that run as separate processes. Two of these are used entirely as provided: a Chroma vector database [47], which stores the visual memory used for retrieval, and local inference servers hosting the vision-language model Qwen3.5-4B [54], which is used for object labelling. The third, Hydra [7], serves as the spatial backbone and builds the layered

3D scene graph with its mesh, object, place, and room layers. Hydra is likewise used largely as provided; its mapping, place extraction, room detection, and loop-closure optimisation are unchanged. It was only configured with a label space of numbered slots instead of semantic classes and minimally modified to support three specific functions of the framework: objects that are segmented only after several frames can still form their own object nodes, the map can be saved and restored across sessions, and incoming frames are discarded rather than queued without limit when reconstruction falls behind. These modifications are described in the sections concerned. Because all components communicate through ROS 2 or standard network interfaces, the framework can be connected to the sensors, localisation, and mapping components of a robot in the same way as any other ROS 2 software.

Conceptually, the framework operates on two time scales. A per-frame path processes every incoming observation at the rate of the camera and is responsible for geometry, tracking, and mapping. An asynchronous semantic path processes individual objects rather than frames and is responsible for labels, confidence values, and mobility. The two paths are connected only through numeric slot IDs that identify persistent objects. The remainder of this subsection follows the data through both paths, and Table **??** summarises the information exchanged between the components.

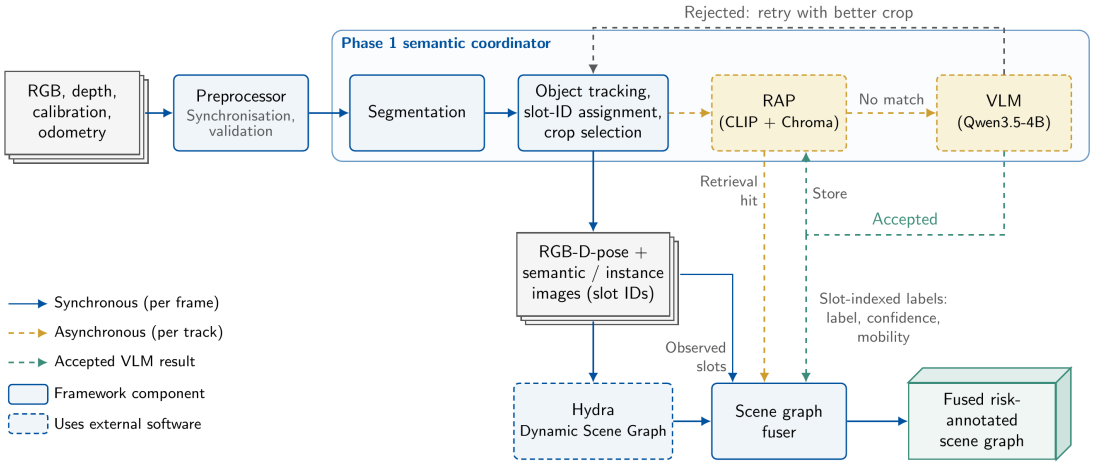


Figure 3.1.: Overview of the proposed framework, showing the synchronous per-frame path (solid) and the asynchronous per-track path (dashed).

Data enters the framework at the preprocessor, which receives RGB images, aligned depth images, camera calibration, and odometry as independent timestamped streams. It associates these streams in time, rejects frames for which no consistent association exists or whose depth data is largely invalid, converts depth into metric units, and expresses the camera pose in the common map frame. The result is a single RGB-D-pose frame in which all components refer to the same instant of observation (Section 3.2). Since every downstream component consumes this representation, synchronisation is resolved once at the entry of the framework instead of repeatedly in each component.

On the per-frame path, the Phase 1 semantic coordinator receives each prepared frame. The colour image is segmented into class-agnostic object masks, which are back-projected with the depth image into 3D point sets with centroids and bounding boxes in the map frame. These observations are associated with persistent object tracks using primarily 3D evidence, such as the position, extent, and accumulated footprint of a track, while overlap in the image is used only as supporting evidence (Section 3.3). An observation that matches no existing track creates a new one. Extended structures such as floors and walls, which can span an entire room, are divided into local segments, so that a single track may be represented by several spatially bounded segments.

Each new track or segment is immediately assigned a slot ID from a fixed range of numbered slots. The coordinator renders the current masks into a semantic image and an instance image whose pixel values are these slot IDs and forwards them to Hydra together with the RGB-D-pose frame. Hydra is configured with this slot range as its label space instead of a fixed set of semantic classes. Since Hydra clusters mesh vertices of the same class into object nodes (Section 2.5.3), and each slot acts as its own class, every Hydra object node corresponds to exactly one track or segment. Hydra therefore receives a fully labelled input stream at the frame rate, even though no semantic label is yet known for a newly observed object, and its object layer inherits the persistent identities established by the tracker.

In parallel, the asynchronous semantic path assigns labels to the tracks. In contrast to the per-frame path, it operates on tracks rather than on frames. While a track is observed, the coordinator collects image crops of the object over a settling interval and retains the crop with the highest quality score, which considers, for example, the size, completeness, and sharpness of the view. Once the track is eligible, its identifier, not its crop, is placed in a queue. The crop is taken from the track only when the identifier is dequeued, so that a better view arriving in the meantime is still used. Labelling is retrieval-first: the crop is embedded with CLIP [45] and compared with the visual memory in Chroma, following the retrieval-augmented perception principle of Section 2.9.1. If a sufficiently similar reference exists, its stored label, confidence, and mobility class are reused. Otherwise, the crop is passed to the VLM in a form that emphasises the target object and suppresses its surroundings, and the model returns a structured result consisting of a label, a confidence value, and a mobility class (Section 3.4). Results that are malformed or fall below a confidence threshold are rejected, and the track waits for a better crop before it is retried a limited number of times. Every accepted VLM result is written back into the memory together with the crop that produced it. Labelling results are published as slot-indexed events. In addition, the coordinator publishes for every frame which slots were observed, which provides the evidence for the presence model of the fuser.

Two properties of this path are essential. First, the VLM only ever receives image crops of a single object; it never receives the full scene graph or a cutout of it, and it is never asked to generate graph structure. Second, retrieval and labelling lie entirely off the per-frame critical path. Their results arrive later and in arbitrary order, which affects only when semantic information becomes available, not whether segmentation, tracking, and mapping can proceed.

The two paths are brought together in the scene graph fuser, which subscribes to Hydra's Dynamic Scene Graph and to the events of the semantic path. Incoming events only update a cache indexed by slot ID; at a bounded rate, the fuser applies the cached labels, confidence values, and mobility

classes to every Hydra object node that carries the corresponding slot (Section 3.5). Labels that arrive long after an object was first mapped are thus applied as soon as they are available. The fuser further completes the relational structure of the graph. Contact between objects and the relation between an object and the local segments of an extended structure are derived from the 3D bounding boxes, room and place membership are taken from Hydra's layers, and missing object-to-place links are added only where a geometric check against the reconstructed mesh confirms them. For every slot, the fuser maintains a presence confidence that is reset by observations and decays while the object is unobserved, at a rate that depends on its mobility class (Section 3.5.2). The result is published as the fused risk-annotated scene graph, which can be visualised in RViz and consumed by downstream risk assessment.

Two further mechanisms keep the scene graph consistent over time. When the localisation back-end corrects accumulated drift, the transform between the map frame and the odometry frame changes. Hydra corrects its own scene graph through its deformation-based optimisation (Section 2.5.4). The coordinator detects such corrections, shifts its cached object geometry accordingly, and merges tracks that were created twice for the same object around the closure, so that the tracker remains consistent with Hydra's corrected map. In addition, both the tracker state and Hydra's map can be saved at the end of a session and restored at the beginning of the next. Re-observed objects are then associated with their earlier tracks in Phase 1 and merged with their restored object nodes in Hydra, so that an environment that is revisited in a later session is not mapped a second time (Section 3.6).

3.1.2. Central Design Decisions

The architecture described above is the result of a small number of design choices, which are best understood as answers to a single underlying problem. The framework has to combine two kinds of processing that operate on very different time scales. Spatial mapping must keep pace with the camera, since a map that lags behind the robot is of little use for assessing the current situation. Open-vocabulary identification, on the other hand, relies on a vision-language model whose inference takes far longer than the interval between two frames. The baseline framework coupled both in a single per-frame loop, so that the slowest component determined the update rate of the entire scene graph. The design choices discussed in this subsection all follow from the aim of separating these two time scales without losing the connection between them.

The starting point was the question of where the spatial representation should come from. The baseline maintained its own frame-based graph, which had neither a common map frame nor loop closure. Extending this graph with the required spatial capabilities would have meant re-implementing incremental mapping, a mesh-grounded object layer, places and rooms derived from free-space topology, and loop-closure correction, all of which Hydra already provides in a real-time architecture that has been evaluated on robotic data [7]. Other spatial scene graph systems were less suitable as a foundation: SceneGraphFusion [22] is restricted to a closed set of classes, and ConceptGraphs [8] applies foundation models to every observation, which is precisely the coupling that limits the update rate. Hydra was therefore adopted as the spatial backbone and deliberately kept close to its original form. Its mapping and optimisation algorithms are used unchanged, and the few modifications are confined to what the framework strictly needs.

This lets the work of this thesis concentrate on what Hydra does not offer, namely open-vocabulary semantics, mobility, and uncertainty, and keeps the framework compatible with future versions of Hydra.

Adopting Hydra, however, brings one of its assumptions with it. Hydra forms object nodes by clustering mesh vertices that share a semantic label, and therefore expects every pixel to be labelled before it is integrated into the map. Supplying real class labels would have required either a closed set of classes, which contradicts open-vocabulary identification, or holding back Hydra's input until the vision-language model has answered, which reintroduces the coupling that the framework is meant to remove. The slot IDs resolve this conflict. Instead of a label, each pixel carries the identifier of the persistent track it belongs to, which is available immediately after tracking. Hydra can thus cluster objects exactly as designed, at its native rate, while never seeing a semantic label at all. The meaning of each slot is supplied later and independently, and because a slot never changes once assigned, the object nodes in Hydra's map inherit the persistent identities established by the tracker.

Once geometry no longer depends on semantics, semantic processing is free to run at its own pace. Labelling is therefore organised around tracks rather than frames and is executed from bounded queues that are decoupled from the camera. A slow or temporarily unavailable inference server can delay labels, but it cannot stall segmentation, tracking, or mapping, and when the queue is full, pending work is deferred rather than allowed to block the pipeline. A useful side effect is that the crop of an object is only chosen when its track is actually processed, so that time spent waiting can produce a better view of the object. The cost of this arrangement is that a newly observed object appears in the map before it has a label. This is acceptable, since such objects are explicitly marked as unlabelled rather than misrepresented, and since they are already geometrically present and can be treated as obstacles by downstream components.

Decoupling makes the cost of the vision-language model tolerable, but it does not reduce it, nor does it make the output of the model any more reliable. Both concerns are addressed in how labelling itself is organised. In a typical indoor environment, the same objects are observed again and again, and many objects resemble others that have already been labelled. Consulting a visual memory before the model resolves such cases at a small fraction of the cost, and because every accepted result is added to that memory, the framework relies on the model less the longer it operates. When the model is needed, it is given a single-object crop and must answer in a fixed format with a small number of fields, each of which is checked before the result is accepted. Malformed or low-confidence answers are rejected instead of being written into the graph. This replaces the free-form graph updates of the baseline, whose errors passed directly into the scene representation, with an output that can be validated while remaining open-vocabulary. Since an incorrect entry in the memory would be propagated to every similar object, only validated results are stored there.

The last choice concerns what happens when semantic information reaches the scene graph. Because labels arrive late and objects move out of view, the fused graph has to express how much it knows about each object rather than present every entry as equally certain. The fuser therefore never alters Hydra's scene graph and attaches all semantic and confidence information as derived metadata, so that the spatial structure remains exactly as Hydra built it. Relationships between objects are derived from their 3D geometry rather than generated by the model, which makes

them properties of the scene instead of impressions that depend on the viewpoint, such as *in front of* or *behind*. This also reduces the load on the vision-language model compared with the baseline, in which the model had to infer all spatial relationships of the visible scene in every inference step, in addition to identifying the objects themselves. In the proposed framework, the model is only asked to identify a single object, which keeps both its input and its output small. Where the baseline deleted objects once their appearance score fell below a threshold, the proposed framework expresses doubt about an object's continued presence through a confidence value that is reset whenever the object is observed and decays while it is not. The rate of decay follows the mobility of the object: the confidence of a static object that is out of view decreases slowly, whereas the confidence for humans, animals, and mobile robots decreases considerably faster, reflecting that they have probably moved. The decay rates are configurable and can be adapted to the dynamics of the environment. Only tracks confirmed as dynamic are eventually withdrawn from the fused output, and the underlying map is never pruned. Downstream risk assessment can thus distinguish objects whose state is well known from those whose state has become uncertain, which is the information that the appearance score of the baseline could not provide.

Taken together, these choices assign each part of the problem to the component best suited to it. Hydra provides spatial consistency and scalability, the slot IDs connect geometry and semantics without making either wait for the other, asynchronous and retrieval-first labelling keep the cost of open-vocabulary identification compatible with near-real-time operation, and confidence-annotated fusion makes the remaining uncertainty explicit. Because all components communicate through ROS 2 or standard network interfaces, the same structure can be integrated into a robotic system without changes to the components themselves.

3.2. Data Acquisition and Preprocessing

The components described in Section 3.1 require each observation to combine visual, geometric, and pose information that refers to the same instant. The sensor data, however, consists of independent timestamped streams of colour images, depth images, odometry, and camera calibration, which are produced at different rates. The preprocessing stage combines these streams into a single *prepared RGB-D-pose frame*, which constitutes the input contract of the framework. Since every downstream component consumes this frame, synchronisation is resolved once at the entry of the framework rather than separately in each component. The stage processes each observation as a sequence of steps, illustrated in Figure 3.2, in which a frame is passed on only if it satisfies both temporal and geometric consistency and is rejected otherwise.

Each frame is anchored to an RGB image, whose timestamp t serves as the reference time for all subsequent steps. The first step checks whether a depth image with approximately the same timestamp is available. If no such depth image exists, the frame is rejected, since the colour image cannot be related to the 3D structure of the scene. The second step checks geometric consistency between the two images, which must have identical resolution, so that each colour pixel corresponds to the same depth pixel and a segmentation mask computed on the RGB image can be lifted directly to 3D. A frame that violates this condition is likewise rejected. The depth

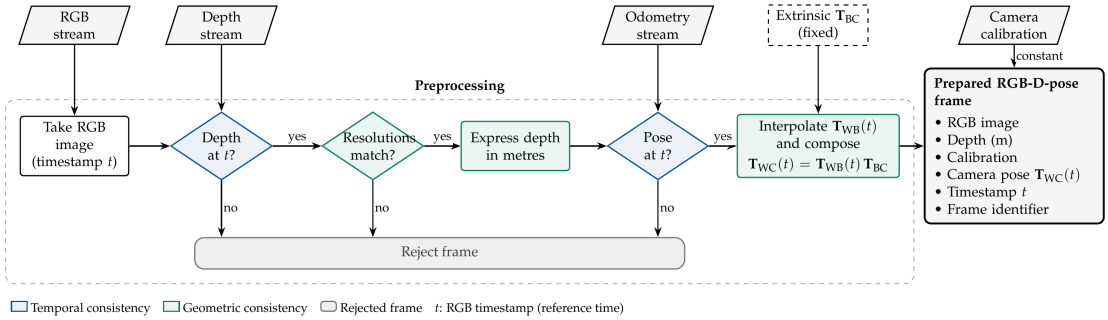


Figure 3.2.: Preprocessing of the input streams into a prepared RGB-D-pose frame.

image is then expressed in metric units, which are used throughout the framework; sources that provide depth in other units are converted at this stage. Depth values are, however, not filtered here, since range filtering is performed in Phase 1 (Section 3.3), where depth is first used.

The third step checks whether the pose of the robot at time t can be determined from the odometry. Since odometry is generally not sampled at the same instants as the camera, the base pose $\mathbf{T}_{WB}(t)$ is interpolated to the RGB timestamp, which reduces the geometric error that would otherwise arise when an image captured during motion is projected with a pose from a slightly different time. If no sufficiently close odometry is available, the frame is rejected. This reflects a deliberate design choice: Hydra [7] integrates every received frame into a persistent map, so discarding an inconsistent frame is preferable to integrating geometrically inconsistent data. Since the odometry describes the pose of the robot base rather than that of the camera, the camera pose is obtained by composing the interpolated base pose with the fixed transform from the robot base to the camera:

$$\mathbf{T}_{WC}(t) = \mathbf{T}_{WB}(t)\,\mathbf{T}_{BC}, \tag{3.1}$$

where $\mathbf{T}_{WB}(t) \in SE(3)$ denotes the pose of the robot base in the world frame at time t, $\mathbf{T}_{BC} \in SE(3)$ denotes the calibrated extrinsic transform from the robot base to the camera, and $\mathbf{T}_{WC}(t)$ is the resulting camera pose. The world frame is the fixed reference frame of the odometry source. Since $\mathbf{T}_{BC}$ is constant, the camera is assumed to be rigidly mounted on the robot.

Finally, a frame that has passed all checks is assembled into the prepared RGB-D-pose frame. It contains the RGB image, the depth image in metres, the intrinsic camera calibration, the camera pose $\mathbf{T}_{WC}(t)$, the timestamp t, and a unique frame identifier. The calibration is not part of the sequential checks, as the intrinsic parameters remain constant during operation and are attached to every frame.

The preprocessing stage is kept intentionally minimal, as it lies on the per-frame path of the framework and any latency it introduces delays all subsequent components. It therefore performs only the operations needed to establish the consistency conditions described above. Furthermore, the method is independent of the data source: applying it to the simulated uHumans2 dataset [11] or the real-world OpenLORIS-Scene dataset [12] changes only the sensor-specific parameters, such as the base-to-camera transform and the depth unit, while the sequence of checks and Equation (3.1) remain unchanged .

3.3. Object Segmentation and Persistent Tracking

Class-agnostic segmentation of the frames prepared in Section 3.2 yields an unordered set of masks without identity. For each mask, the framework must decide whether it shows a new physical object or a further observation of a known one, and it must maintain a persistent three-dimensional estimate of every object. The two association errors identified in Section 2.8.1 have strongly asymmetric consequences here. If two objects are merged into one track, its geometry describes neither of them and the crop passed to the vision-language model shows parts of both, so the label is wrong for both; since label and geometry belong to the track as a whole, this cannot be repaired downstream. If one object is fragmented into several tracks, the representation becomes redundant, but each fragment remains correct and can, in principle, be merged later. The method therefore follows a single design principle: whenever the evidence is ambiguous, fragmentation is preferred over over-merging.

3.3.1. Object Proposals and 3D Observations

Object proposals are generated by a promptable model of the Segment Anything family [30], prompted with a regular grid of foreground points. A common interface returns a set of class-agnostic masks, so the remainder of the method is independent of the model behind it. Two interchangeable models are supported: the original SAM with a ViT-B encoder and NanoSAM (Section 2.7), which has a considerably lower latency on embedded hardware but reduced accuracy, particularly for small objects [33]. SAM uses its own automatic mask generation, whereas for NanoSAM the equivalent post-processing is realised as a wrapper (Section 2.7.4). In both cases, duplicate masks from neighbouring prompts are removed by non-maximum suppression, and implausibly small masks are discarded, as they rarely represent complete objects and provide too few depth measurements. To bound the cost of all subsequent stages, the number of proposals per frame is limited, retaining the masks with the highest predicted quality. Since the choice of model trades proposal quality against update rate, it is determined experimentally in Chapter **??**. The result is a set of masks $\mathcal{M}_t = \{m_1, \ldots, m_N\}$ for frame t.

Each mask is converted into a three-dimensional observation. Depth is taken only from the largest connected region of the mask, because masks often contain detached fragments and boundary regions whose depth belongs to the background. A pixel (u, v) with valid depth d is back-projected with the pinhole model and transformed into the map frame using the camera pose of Equation 3.1:

$$\mathbf{p}^c = d \begin{bmatrix} (u - c_x)/f_x \\ (v - c_y)/f_y \\ 1 \end{bmatrix}, \qquad \mathbf{p}^m = \mathbf{R}_c^m \, \mathbf{p}^c + \mathbf{t}_c^m, \tag{3.2}$$

where f_x, f_y, c_x and c_y are the focal lengths and principal point, and $(\mathbf{R}_c^m, \mathbf{t}_c^m)$ is the transformation from the camera to the map frame. The resulting point set $\mathcal{P}_i$ is summarised as an observation $o_i = (\mathcal{P}_i, \mathbf{c}_i, \mathcal{B}_i, \mathbf{b}_i)$ with the axis-aligned three-dimensional bounding box $\mathcal{B}_i$ of the points in the map frame, its centre $\mathbf{c}_i$, which serves as the centroid of the observation, and the two-dimensional

bounding box $\mathbf{b}_i$ of the mask in image coordinates. Since thin structures such as floors, ceilings and walls yield boxes of almost zero thickness, for which a volumetric overlap is not meaningful, every box dimension below a minimum thickness is padded to that thickness before $\mathbf{c}_i$ and the volume are computed. The three-dimensional quantities provide viewpoint-invariant evidence for association, while $\mathbf{b}_i$ is used for the image-space comparison and for extracting the object crop. A mask without reliable geometry, for instance because most of its depth values are invalid, is discarded for the current frame, and no observation, crop or track is created from it. The object is not ignored: it is expected to be observed again with meaningful depth as the robot moves, and is tracked then. Discarding the mask only keeps inaccurate data out of the persistent representation; moreover, an observation without geometry could not be associated by the three-dimensional evidence and would only open short-lived tracks.

Depth values outside a valid sensing range are excluded during back-projection, completing the filtering deferred in Section 3.2. This serves two purposes. First, depth accuracy decreases with distance: for a stereo camera, the error grows approximately quadratically, $\delta z \approx z^2 \, \delta_d / (f \, b)$, with depth z, focal length f, baseline b and disparity error δ_d [60]. Distant points therefore yield noisy centroids and inflated boxes, which weaken the association cues and, through the accumulated extent (Section 3.3.3), could permanently distort a track. Measurements below the sensor's minimum operating distance are excluded for the same reason. Second, the restriction limits the computational cost per frame, since masks of distant objects are discarded before association, crop selection and labelling, so that the expensive vision-language model inference is applied only to objects near the robot. Distant objects are observed again once they enter the valid range, and nearby objects are generally the most relevant for immediate risk assessment.

3.3.2. Evidence-Based Association

Association is performed primarily in three dimensions, because the position and extent of a static object in the map frame are approximately constant across viewpoints, whereas its image appearance is not (Section 2.8.1). Fusion++, in contrast, relies on a single image-space overlap criterion [43]. An observation o is compared only with candidate tracks τ in its spatial neighbourhood, which are retrieved by a grid search over a uniform grid in the ground plane: each track is registered in the cells covered by its accumulated extent and by its centroid, and the candidates are the tracks registered in the cells covered by the observation's footprint or lying within the centroid acceptance radius. Since a track outside these cells can pass neither the overlap nor the centroid test, it cannot satisfy the acceptance condition below, so the grid search reduces the cost per observation without changing the association result. The evidence is organised hierarchically: the three-dimensional overlap is the primary evidence, while centroid distance and image overlap are secondary evidence that is consulted only when the overlap is not conclusive.

The primary evidence is the volumetric overlap between the observation's bounding box and the accumulated extent of the track, which reflects that a static object occupies the same region of the map from any viewpoint. The accumulated extent is used because successive views often capture different parts of an object, and overlap is required on both horizontal axes, so that boxes that merely touch do not count. Two levels are distinguished. A partial overlap shows a shared

region but is ambiguous, since adjacent objects also overlap partially. A near-complete inclusion of the observation in the track's extent, referred to as containment, is the expected outcome of a re-observation, especially for large objects such as floors and walls that are seen only in part. In addition, a height test requires only the vertical extents to overlap or to be separated by at most a small gap, irrespective of horizontal position. A volumetric overlap therefore implies height compatibility, but not vice versa, since boxes at the same height may lie side by side.

The secondary evidence resolves the ambiguity of a partial overlap. The centroid distance separates objects whose boxes intersect but whose centres lie apart. It is mapped to a similarity by a Gaussian kernel whose width scales with object size, because the centroid of a large, partially observed object shifts more between views. The image overlap between $\mathbf{b}$ and the box of the track's last observation is independent of depth and pose errors and strongly indicates continuity between consecutive frames, but its value decreases quickly as the viewpoint changes.

Each test k yields a similarity $\sigma_k(o,\tau) \in [0,1]$ and a binary verdict $\pi_k(o,\tau) \in \{0,1\}$. With π_{con}, π_{ov}, π_{h}, π_{cen} and π_{img} denoting the verdicts for containment, partial overlap, height, centroid distance and image overlap, a match in association mode μ is accepted if

$$
\begin{aligned}
s^{(\mu)}(o,\tau) = \frac{\sum_k w_k^{(\mu)}\, \sigma_k(o,\tau)}{\sum_k w_k^{(\mu)}} \geq s_{\min}^{(\mu)} \quad \wedge \quad \neg\, \chi(o,\tau) \quad \wedge \\
\begin{cases}
\text{no further condition} & \text{if } \pi_{\text{con}} = 1, \qquad \text{(Tier 1)} \\
\pi_{\text{cen}} \vee \pi_{\text{img}} & \text{if } \pi_{\text{ov}} = 1 \text{ and } \pi_{\text{con}} = 0, \qquad \text{(Tier 2)} \\
\pi_{\text{h}} \wedge \pi_{\text{cen}} \wedge \pi_{\text{img}} & \text{otherwise,} \qquad \text{(Tier 3)}
\end{cases}
\end{aligned}
\tag{3.3}
$$

where $w_k^{(\mu)}$ are mode-dependent weights with the largest weight on the volumetric overlap, and the score rejects matches with only marginal support. Under containment (Tier 1), the overlap alone decides, based on the assumption that a near-complete inclusion in a track's extent indicates a re-observation of the same object; the cases in which this assumption fails, such as an extent enlarged by an earlier erroneous association or a small object stored inside a larger one, are examined in the evaluation. Under partial overlap (Tier 2), one secondary cue must confirm the match, and otherwise (Tier 3), both secondary cues and the height test must agree. Tier 3 mainly serves young tracks of small objects. Depth errors displace points along the viewing ray [60], which is approximately horizontal for a forward-facing camera, so the observed box is shifted mainly in the ground plane. Because the volumetric overlap is multiplicative across axes, small horizontal shifts can prevent a sufficient overlap for the same object, while its vertical extent is barely affected. The height test is therefore reliable in this situation, and the agreement of both secondary cues guards against horizontally adjacent objects. An image-only match that contradicts the three-dimensional evidence, such as an object in front of a wall, falls into Tier 3 and fails on the centroid. Since containment implies partial overlap, which implies height compatibility, the rule is equivalent to a quorum of three agreeing verdicts among the five tests, which is how it is implemented. Finally, the predicate χ rejects a match whose observation volume is grossly inconsistent with the smoothed volume of the track; as this reference is formed from individual observations, χ cannot detect an extent that has grown through an erroneous association.

Two association modes depend on the time $\Delta t = t - t_\tau$ since the track was last observed. In the recent mode, the viewpoint has changed little and image overlap is trusted. In the revisit mode,

the object is seen again after a longer absence, possibly from a very different viewpoint, so much stronger image agreement is required and centroids are compared in the ground plane only, since the height of a partially observed object's centroid depends on the visible part.

Before any track is updated, all observations of a frame are evaluated against the track states at the beginning of the frame. Segmentation often emits a merged region together with its parts, so a mask largely covered by smaller masks already matched to distinct tracks is suppressed. Nested masks that are admissible for the same track describe the same visible surface twice; only one of them is retained, where a broader mask replaces a nested part if it adds new visible area with consistent three-dimensional support, so that the track can grow as more of the object becomes visible. Each remaining observation o_i is then assigned independently to its admissible track with the highest association score,

$$\tau^*(o_i) = \arg \max_{\tau_j \in \mathcal{E}(o_i)} s^{(\mu)}(o_i, \tau_j), \tag{3.4}$$

where $\mathcal{E}(o_i)$ denotes the set of tracks for which o_i satisfies Equation 3.3; if $\mathcal{E}(o_i)$ is empty, o_i starts a new track. Since all observations are assigned against the same frame-initial track states, the assignment does not depend on the order of the segmentation output. Unlike a one-to-one assignment, several non-nested observations may be assigned to the same track within one frame, so that an object divided into adjacent masks is not fragmented; these observations are integrated successively into the track and its local segments (Section 3.3.3). As a consequence, competition between observations does not prevent over-merging: an observation admissible for several tracks is assigned to the highest-scoring one irrespective of the other observations, and a new object admissible for an adjacent known track is absorbed into it. Protection against over-merging therefore rests entirely on the conservative acceptance condition of Equation 3.3.

3.3.3. Track Representation and Local Segments

Each track holds a persistent identity, a smoothed centroid and volume, an accumulated three-dimensional extent, and the image-space bounding box of its last observation. Centroid and volume are updated as exponential moving averages, for instance $\mathbf{c}_\tau \leftarrow \alpha\, \mathbf{c}_\tau + (1 - \alpha)\, \mathbf{c}_o$ with $\alpha \in (0, 1)$, which damps per-frame noise. The accumulated extent is the smallest axis-aligned box enclosing all associated observation boxes. It grows monotonically and thus approaches the full object over several partial views, but it never shrinks, so an erroneous association enlarges it permanently and facilitates further ones. The track centroid is deliberately not taken as the centre of this extent. Since an observation usually covers only part of an object, its centroid is the centre of the visible part, whereas the centre of the extent is the centre of everything observed so far. For a large object observed while the robot moves along it, the distance between the two grows with the distance travelled, so that a correct observation would fail the centroid test. Consecutive observations, in contrast, show largely the same part of the object, and the moving average follows them with a lag of approximately $\Delta c / (1 - \alpha)$, where Δc denotes the displacement of the observed centroid between frames, independently of the size of the object. In addition, an outlier observation affects the moving average only temporarily, whereas it would enlarge the extent, and thus shift its centre, permanently. This compact state requires constant memory per track.

Extended structures such as floors, walls and ceilings cannot be represented by a single box, since its extent would cover a large part of the map and its centroid would lie far from most of the structure. Each track is therefore divided into local segments of bounded horizontal span. Every segment receives its own slot ID (Section 3.1) and thus its own object node in Hydra [7], while the track retains one identity and therefore one label for all its segments. An observation extends an open segment as long as the span limit is respected. A segment that reaches the limit is closed; subsequent observations either re-attach to it without enlarging its geometry or open a new segment. A single mask that already exceeds the limit is not split.

3.3.4. Transfer of Object Observations to Hydra

The tracking results of every frame are passed to Hydra [7] independently of the semantic labelling, so that the spatial representation is built up without delay. The masks are rendered into a semantic image and an instance image (Section 3.1.1), in which each pixel carries the slot ID of the assigned local segment, so that Hydra forms one object node per segment. Each mask is rendered in the filtered form from which its geometry was computed, and overlapping masks are drawn in order of decreasing area, so that every pixel carries a single slot ID and small objects remain visible in front of large structures. Pixels without a track or with depth outside the valid range receive a reserved background ID and invalid depth, and for real sensor data the slot ID of a new track can be withheld until it has been observed in several frames. All data are transmitted with the timestamp of the original observation, so that Hydra integrates geometry and object identity from the same instant.

3.4. Semantic Labelling and Risk-Relevant Object Information

Section 3.1 introduced the asynchronous, retrieval-first organisation of labelling and the principle of validating every result before it is stored. This section describes the two aspects that determine the quality of the semantic information within that organisation: how the view of an object is selected and presented to retrieval and to the VLM, and what the structured result contains beyond the label itself.

3.4.1. Selection and Presentation of the Object View

Since a track is labelled from a single image, the quality of that image largely determines the quality of the label. Each observation of a track is therefore assessed with three no-reference quality cues. The prominence P measures the number of object pixels and is normalised logarithmically, reflecting the diminishing gain in visual information as an object grows larger in the image. The sharpness S_{sharp} is estimated from the variance of the Laplacian response, a widely used focus measure [61]. The framing M expresses how little of the object mask touches the border of the

crop, and thus penalises objects that are cut off by the image boundary. With all cues normalised to $[0, 1]$, the score of a candidate and the condition under which it replaces the stored crop are

$$S = w_P\, P + w_S\, S_{\text{sharp}} + w_M\, M, \quad w_P + w_S + w_M = 1, \qquad S_{\text{new}} > (1 + h)\, S_{\text{old}}, \tag{3.5}$$

where w_P, w_S and w_M weight the cues and h is a relative improvement margin.

The cues are combined additively rather than multiplicatively. In a product, a single weak cue suppresses the entire score, so that dissimilar candidates collapse to similarly low values; a sum preserves the contribution of the remaining cues and keeps candidates distinguishable. The margin h introduces hysteresis into the selection. Because view selection continues for as long as a track waits to be processed, consecutive frames supply many near-identical views whose scores differ only by noise; requiring a clear improvement prevents their continual exchange while still admitting genuinely better views as the robot approaches an object.

Once a track is processed, the selected view is presented differently to retrieval and to the VLM. Retrieval receives the crop defined by the bounding box of the object, without further modification. The VLM receives a crop in which this bounding box is enlarged by a margin of surrounding context, so that the object can be interpreted in relation to its surroundings and parts of it that the segmentation mask has missed remain visible.

Within this enlarged crop, the target is marked by a boundary. The boundary is drawn along the outer contour of the segmentation mask rather than along the bounding box, and is rendered as a thin line in a conspicuous colour, while the pixels of the object itself remain unaltered. Since an enlarged crop frequently contains several objects, the boundary specifies which of them is to be named. The instruction given to the model refers explicitly to this boundary and states that everything outside it serves only as context, so that the model identifies the marked region rather than the most salient object in the crop.

3.4.2. Structured Output and Risk-Relevant Object Information

The structured result of the VLM comprises five fields: an open-vocabulary label, a label confidence, a mobility class, a mobility confidence, and a short description of the visual state of the object. The mobility class distinguishes three cases. An object is classified as dynamic only if it can change its position by its own means, which applies to humans, animals and self-propelled machines such as mobile robots. All other objects are classified as static, including the structural elements of a room, fixtures and furniture. The distinction thus refers to the capacity for self-propelled motion rather than to whether an object can be moved at all: an object that is occasionally relocated by a person remains static, since it does not change its position unless acted upon. The unknown class is used when the object itself could not be identified, so that no statement about its mobility can be made. An explicit no-result answer is provided for crops in which nothing can be identified, so that the model is not forced to name an object it cannot recognise.

The label states what an object is, whereas its risk relevance frequently depends on its current state. The visual-state description therefore records what is visibly particular about the instance

at the time of observation, namely items placed on or with it, its condition, or whether it appears active or in use, such as water on a floor or a display that is switched on. It excludes identity, material and colour, which are covered by the label, and has an explicit value for the case that nothing notable is visible. Together with the mobility class, this description constitutes the risk-relevant object information that the scene graph provides to downstream risk assessment; the framework records these observations rather than computing a risk score itself. As open-ended text, it is not used to accept or reject a result. When a label is obtained by retrieval, however, the stored description belongs to the reference observation rather than to the current instance, so the description is most informative for objects labelled directly by the VLM.

Validation treats the fields asymmetrically. A result is rejected only on the basis of its label and label confidence, whereas a weak mobility decision is reduced to the unknown class without discarding an otherwise valid label, and a mobility class that contradicts an unambiguous label, for instance a person reported as static, is corrected for consistency. The instruction given to the model is organised as a structured decision procedure with priorities for ambiguous regions, such as surfaces carrying fixtures or transparent partitions, together with explicit guidance on calibrating the reported confidence. Its examples illustrate this procedure rather than enumerate permissible labels. The prompt variants considered during development are compared in Chapter **??**.

3.5. Scene Graph Fusion and Uncertainty Modelling

The semantic interpretation of objects (Sections 3.3 and 3.4) and the spatial scene graph maintained by Hydra [7] are produced independently and at different rates. This section describes how both are combined into a single risk-annotated scene graph (Section 3.5.1) and how the uncertainty associated with each object is represented within it (Section 3.5.2).

3.5.1. Scene Graph Fusion

The fusion stage combines the spatial scene graph of Hydra with the semantic results and observation history of the tracked objects. Its central design principle is that Hydra's graph is treated as the authoritative spatial structure and is never modified. Labels, relations and confidence values are added on top of this structure rather than written into it. They are not kept as a separate record that is updated step by step; instead, they are computed anew from Hydra's graph each time Hydra provides an update. As a consequence, structural changes made by Hydra, for example when object nodes are merged or corrected after loop closure (Section 3.6), are reflected automatically in the fused graph, and semantic information cannot drift out of alignment with the spatial map.

Semantic information is attached to Hydra's object nodes through their common slot. In the normal case, all nodes associated with a slot receive the same label, mobility class, descriptive object properties and associated confidence values. Where Hydra represents a single object by several overlapping nodes that share a slot, these nodes are consolidated into one node in the

fused graph, and their edges are transferred to it, so that each physical object appears only once. Conflicting semantic results for the same slot, which can arise when Hydra merges or corrects nodes, are resolved by spatial proximity, so that each node receives the interpretation that was observed closest to it. Objects that have not yet been interpreted are represented explicitly as such, which allows the graph to distinguish between an object of unknown identity and the absence of an object.

Before relations between objects can be derived, it must be clear which graph nodes belong to the same physical object. A large structure, such as a long wall, may be represented by several local segments (Section 3.3), each of which appears as a separate node in the graph. If relations were derived between these nodes directly, a single physical relationship, such as a table standing against the wall, would appear several times, once for each wall segment. To avoid this, nodes that share a common track identity are first grouped into a *main object*, and relations are always reasoned about at this level.

Two kinds of relations are derived between main objects. The *identity relation* connects the segments of one main object and states that they are parts of the same physical object. It follows directly from the tracking result and does not depend on geometry. The *contact relation* connects two different main objects and states that they touch or nearly touch. Contact is first decided for the two objects as a whole, and the resulting edges are then attached to those segments of the two objects that lie closest to each other. Since identity relations only link parts of the same object, while contact relations only link different objects, the two relation types never connect the same pair of nodes and therefore cannot contradict each other.

The spatial membership of objects and places is taken from Hydra wherever it is available and is completed only where it is missing. Associations between objects and places, and between places and rooms, are added only when the reconstructed geometry of the environment confirms that the two elements are mutually visible, so that no association is created through a wall or across floors. Where such geometric confirmation is not possible, no association is derived.

All derived relations therefore follow the same principle: they are added only when supported by geometric evidence in the shared three-dimensional frame. As a result, they describe properties of the scene itself and remain the same regardless of the viewpoint from which the robot observes it.

3.5.2. Uncertainty and Persistence Modelling

For each object, the framework represents three kinds of uncertainty separately: uncertainty about its semantic identity, about its mobility, and about its continued presence at the last observed position. These quantities answer different questions for downstream risk assessment and behave differently over time: identity and mobility are fixed once an object has been labelled, whereas the belief in its presence changes continuously.

Semantic identity and mobility are determined together, once per tracked object (Section 3.4). If a sufficiently similar object has been labelled before, its label, mobility class and object description are taken over through visual retrieval; otherwise, they are obtained directly from the VLM. Because tracks with conflicting labels are never merged (Section 3.3), each object retains a single label throughout its lifetime, and objects for which no usable label is obtained remain explicitly unclassified. The uncertainty of identity and mobility is represented by the confidence values attached to them, which are carried into the scene graph. For a retrieved label, this confidence is bounded both by the confidence with which the original object was labelled and by the similarity between the two objects. A label transferred from a weakly matching or uncertainly labelled object is therefore marked as correspondingly uncertain, which allows downstream risk assessment to distinguish confidently identified objects from those whose identity is doubtful.

Presence is represented by a confidence $p \in [0, 1]$, which expresses the belief that an object is still located where it was last observed. It is derived from the time of the most recent observation of the object, is equal to one while the object is being observed, and otherwise decays over time at a rate that depends on its mobility class m,

$$p(t) = 0.5^{(t-t_{\text{last}})/\tau_m}, \qquad \tau_{\text{dynamic}} \ll \tau_{\text{static}}, \tag{3.6}$$

where t_{last} is the time of the most recent observation and τ_m the half-life associated with the mobility class. The decay rates can be adapted to the dynamics of a given environment. How the mobility class shapes this decay, and what follows from it, is described in Section 3.5.3.

By separating what an object is, whether it moves, and whether it is still present, the representation enables downstream risk assessment to distinguish well-established objects from uncertain ones. These quantities are, however, confidence values and decay heuristics rather than calibrated probabilities. They inherit the uncalibrated nature of VLM-reported confidence (Section 2.9.2) and should therefore be interpreted as relative measures of reliability.

3.5.3. Treatment of Static and Dynamic Objects

The mobility class determines how an object is represented and maintained over time. An object is classified as *dynamic* only if it can move by itself, as humans, animals and mobile robots do. All other objects, including furniture that is occasionally moved by others, such as chairs, are classified as *static*. If the mobility of an object cannot be determined with sufficient confidence, it is classified as *unknown*. To reduce misclassifications, the mobility class proposed by the VLM is checked against the object category: clear labels of humans, animals or mobile robots enforce the dynamic class, whereas clear labels of structures and furniture enforce the static class (Section 3.4).

The two classes reflect two different assumptions about the environment. For a static object, the default assumption is that it remains where it was observed, even when it has not been seen for some time. For a dynamic object, the default assumption is the opposite: once it is no longer observed, it has most likely moved elsewhere. This difference affects three aspects of the representation, which are summarised in Table 3.1.

Table 3.1.: Differences in the treatment of static and dynamic objects.

Aspect	**Static (and unknown)**	**Dynamic**
Typical objects	Walls, floors, furniture, fixtures	Humans, animals, mobile robots
Spatial representation	Node created by Hydra from the accumulated reconstruction	Most recent observed position; added to the fused graph if Hydra has no node
Presence decay	Slow	Fast
When no longer observed	Retained with decreasing presence confidence	Withdrawn once presence has become unlikely

The first aspect is the spatial representation. Hydra creates object nodes from surfaces that are reconstructed consistently over time [7]. This works well for static objects, whose surfaces accumulate at the same location with every observation. A moving object, by contrast, does not remain in place long enough for its surface to accumulate, so that Hydra may never create a node for it, even though it has been reliably identified by the perception pipeline. Such dynamic objects are therefore added to the fused graph by the fusion stage itself, using only their most recent observation. The extent observed over the whole track is deliberately not used, since for a moving object it would cover the entire path travelled rather than the space the object currently occupies. Without this step, the objects that are most relevant for risk assessment would be missing from the scene graph precisely because they move.

The second aspect is the decay of the presence confidence (Equation 3.6). Because a dynamic object is expected to change its position, its presence confidence decays within a short time after it was last observed. The presence confidence of a static object decays slowly, so that furniture and structures remain credible long after leaving the field of view. Objects of unknown mobility are treated as static, since in the absence of evidence that an object moves, assuming that it has stayed in place is the safer choice.

The third aspect concerns what happens to an object that is no longer observed. A dynamic object is withdrawn from both the tracking state and the fused graph once its presence confidence has fallen below a threshold, since it has then most likely left its last observed position. This prevents outdated positions of people or robots from suggesting hazards where none exist. Static objects and objects of unknown mobility are never withdrawn. They remain in the scene graph with a correspondingly low presence confidence, so that downstream risk assessment can still take them into account while recognising that their position has not been confirmed recently. In no case is an object removed from Hydra's map. This asymmetry is deliberate: for risk assessment, wrongly removing a static obstacle that is still present is more harmful than retaining an object whose presence is uncertain.

3.6. Loop Closure and Multi-Session Consistency

The tracks of Phase 1 cache object geometry in the map frame and must remain consistent with Hydra's map whenever that map changes, whether through a drift correction within a session or through a restart between sessions. Otherwise, a re-observed object no longer matches its own track and creates a duplicate instead of being re-identified.

3.6.1. Consistency under Loop Closure

When the localisation back-end corrects accumulated drift, Hydra deforms its scene graph accordingly (Section **??**). The cache of the tracker is not part of this optimisation and would remain in the uncorrected frame. Re-observed objects would then fail the association tests of Section 3.3.2 and start new tracks, so that Phase 1 diverges from Hydra.

A correction becomes visible as a change in the transform $\mathbf{T}_{\mathrm{MO}}$ between the map frame and the odometry frame, and changes below a small threshold are ignored as jitter. This change, however, describes the correction at the current pose only. Since drift accumulates along the trajectory, geometry observed shortly before the closure carries almost the full error, whereas geometry observed at the beginning of the loop carries almost none. Applied rigidly to the entire cache, the change would be exact for the most recent objects but would displace the earliest ones, which are precisely those re-observed when the loop is closed. Hydra's optimisation, in contrast, distributes the correction along the trajectory and moves every object node individually.

The tracker therefore adopts Hydra's correction instead of estimating its own. Since every local segment carries its own slot ID and thus corresponds to exactly one object node (Section 3.3.3), the correction of a segment can be read directly from the movement of its node. Only segments for which Hydra has not yet created a node, typically those observed shortly before the closure, require a different treatment. Every point $\mathbf{x}$ of a segment s, that is, its centroid and the corners of its bounding box, is therefore moved to

$$\mathbf{x}' = \begin{cases} \mathbf{x} + (\mathbf{p}'_s - \mathbf{p}_s) & \text{if Hydra has a node for } s, \\ \mathbf{T}'_{\mathrm{MO}}\, \mathbf{T}_{\mathrm{MO}}^{-1}\, \mathbf{x} & \text{otherwise,} \end{cases} \tag{3.7}$$

where a prime denotes a quantity after the correction. The first case applies to segments that Hydra has already integrated into its graph. Here, $\mathbf{p}_s$ and $\mathbf{p}'_s$ are the positions of the node of s in the last graph before and the first graph after the correction, so that their difference is exactly the displacement that Hydra's optimisation has assigned to this object. Since both positions are expressed in the map frame, this displacement already contains the complete correction of the object, and the change in $\mathbf{T}_{\mathrm{MO}}$ is not applied in addition. Adding it to $\mathbf{x}$ moves the cached segment by the same amount as its node, and the tracker therefore inherits Hydra's non-rigid correction segment by segment without estimating any correction of its own.

The second case applies to segments without a node and is read from right to left. A cached point was originally obtained in the odometry frame and converted into the map frame with

the transform valid at that time, $\mathbf{x} = \mathbf{T}_{\mathrm{MO}}\,\mathbf{x}_{\mathrm{O}}$, where $\mathbf{x}_{\mathrm{O}}$ denotes its odometry coordinates. A loop closure changes only this transform, not the odometry measurement itself. Multiplying by $\mathbf{T}_{\mathrm{MO}}^{-1}$ therefore recovers $\mathbf{x}_{\mathrm{O}}$, and multiplying by $\mathbf{T}'_{\mathrm{MO}}$ converts it into the map frame with the corrected transform, as if the correction had already been known when the point was observed. Both transforms are applied in homogeneous coordinates. Because $\mathbf{T}'_{\mathrm{MO}}$ compensates the drift accumulated up to the current pose, this case is only valid for geometry that carries the same drift, which holds for the recently observed segments to which it is applied.

Tracks that were nevertheless created twice around the closure are folded into their earlier identity by a subsequent merging step. A recently created track is merged into an older compatible track, that is, one without a conflicting label, if it lies within a search radius that scales with the magnitude of the correction, and pairs whose extents overlap strongly are merged as well. The older identity, together with its slot and label, survives. This is the later repair of fragmentation anticipated in Section 3.3.

3.6.2. Persistence across Sessions

A robot that is switched off and restarted should neither forget what it has mapped nor build a second copy of the environment once it resumes operation. This work considers a restart in place, in which the robot is started again where it was switched off, so that its last known pose remains valid. This pose is saved together with the map at shutdown and used as the initial pose of the new session, which therefore continues in the same map frame as the previous one.

The three components that hold state save it independently. The tracker saves its tracks, including their geometry, local segments with their slot IDs, labels and mobility classes, together with its slot and identity counters, so that no identifier is issued twice. Hydra saves its scene graph and mesh, and the retrieval memory retains its stored references (Section 3.4). The fuser holds no state of its own, since its output is recomputed from the other components (Section 3.5.1). After the restart, restored tracks take part in the association like any other track, and their labels are reused directly without querying retrieval or the VLM again.

Hydra can save its map but provides no means of resuming from it, which required three conceptual changes. First, the saved scene graph and mesh are loaded as the initial map of the new session before any new data are integrated. Second, all restored nodes are marked as archived, that is, as lying outside the active window in which Hydra is currently mapping, because Hydra only considers archived nodes as merge candidates for newly mapped objects. Third, the identifiers of new nodes continue after the highest restored identifier, and new geometry is appended to the restored mesh, so that new and restored elements never collide. The trajectory of the previous session is not restored; the new session begins a new trajectory at the saved pose.

Restored nodes are reconciled with newly mapped ones by Hydra's object merging, which is enabled for this purpose and joins nodes that carry the same label and are spatially consistent. Merging by label is normally risky, since many physical instances share a class, but here it is safe:

as a direct benefit of the slot design (Section 3.1.2), labels are per-object slot IDs, so two distinct objects can never share one.

4. Implementation

This chapter describes how the methodology presented in Chapter **??** was realised. Section 4.1 establishes the hardware and data basis of the work, including the data-collection campaign that was originally planned and the reasons why substitute datasets were adopted. Section **??** describes the software implementation of the framework, and Section **??** the measures taken to achieve near-real-time operation on embedded hardware.

4.1. Hardware Platforms and the Go1 Data-Collection Campaign

4.1.1. Compute Platforms

The pipeline was developed, profiled and optimised on an NVIDIA Jetson AGX Orin [56] and subsequently ported to an NVIDIA Jetson AGX Thor [58], on which the final runtime results were obtained. On both platforms, all components of the framework are executed on the same board: the segmentation model, the embedding model and vector database used for retrieval, the local inference server hosting the vision-language model, Hydra [7], and the three framework nodes, namely the preprocessor, the Phase 1 semantic coordinator and the scene graph fuser. The specific models are selected through the configuration and can be exchanged without changes to the code. This configuration corresponds to the intended onboard deployment, but it also means that segmentation, retrieval, VLM inference and volumetric mapping compete for the same GPU and the same unified memory. Contention for these shared resources is therefore a central factor for the achievable update rate and is addressed in Section **??**.

4.1.2. Planned Data-Collection Campaign

The publicly available datasets considered in this work were recorded with wheeled robots, handheld sensors or in simulation. To evaluate the framework on the type of platform for which it is intended, a dedicated real-world dataset was planned, recorded with a legged robot inside a university building. In the intended deployment scenario, the robot carries an RGB-D camera and sufficient onboard computing resources to execute the complete perception, mapping and annotation pipeline during operation.

The planned setup consisted of an Intel RealSense D455 RGB-D camera [62], which provides synchronised colour and depth images, mounted on a Unitree Go1 quadruped robot [63]. The camera was rigidly attached to the robot body by a custom 3D-printed base, and the transform between the robot base and the camera was defined manually as a fixed extrinsic, corresponding to the constant transform $\mathbf{T}_{BC}$ assumed in Section 3.2. The camera was connected via USB to a Raspberry Pi onboard the robot, which ran the RealSense driver under ROS 1 Melodic, in line with the ROS 1-based software stack of the Go1. The colour, depth and calibration messages were published over the internal network of the robot, on which a separate onboard computer subscribed to them together with the robot odometry.

During operation, this computer recorded the colour images and the depth images at a resolution of 640×480 pixels, together with their calibration, the depth-to-colour extrinsics and the robot odometry, as ROS 1 bags directly on the robot. Depth was recorded in its native sensor geometry rather than aligned to the colour image, which kept the processing load on the Raspberry Pi low, while the recorded extrinsics retained the information required for a later alignment. Since the onboard computer did not provide sufficient free storage for RGB-D recordings, the bags were to be written to a USB storage device connected to it. The recordings had to contain complete and temporally consistent streams, since accurate temporal alignment between the images and the odometry is a precondition for the frame preparation described in Section 3.2. The recorded bags were subsequently to be converted into the ROS 2 bag format and replayed through the ROS 2 pipeline [10], with all processing on the Jetson performed offline. For an actual deployment of the complete system, the robot would either have to provide a native ROS 2 interface or use a ROS 1–ROS 2 bridge that transfers the sensor and odometry data into the ROS 2 pipeline during operation.

The campaign was carried out but ultimately discarded, since the limitations described in Section 4.1.3 prevented the recording of data suitable for the evaluation. The framework was therefore evaluated on publicly available datasets instead (Section 4.1.4). The office sequences of the real-world OpenLORIS-Scene dataset [12] replace the planned recordings, as they were captured with a RealSense camera of the same sensor family as the D455. They are complemented by two simulated scenes of the uHumans2 dataset [11], an office and an apartment. The apartment sequence contains a human and thus provides a dynamic object, which allows the mobility classification and the mobility-dependent presence model of Section 3.5.2 to be examined.

4.1.3. Limitations Encountered

Several limitations arose during the campaign, which are summarised in Table 4.1. They fall into three categories: the integration of the compute platform with the robot, the recording of the data, and the quality of the depth measurements.

The first category concerns platform integration. The Go1 provides neither a suitable power supply nor a hardware interface through which the Jetson AGX Thor could be operated onboard, so that the framework could only be applied offline after recording (limitation 1). This restricts the campaign to offline evaluation, but does not prevent it.

Table 4.1.: Limitations encountered during the Go1 data-collection campaign.

Limitation	Cause	Consequence for this thesis
(1) Onboard compute	No suitable power supply or hardware interface on the Go1 to operate the Jetson AGX Thor onboard	Processing only possible offline after recording
(2) Recording bandwidth	Only USB 2.0 ports on the robot for camera and storage; D455 restricted to 5, 15 or 30 fps at 640 × 480	Delivered rates below 15 fps and frame loss; no reliably complete, temporally consistent recordings
(3) Vibration-induced depth noise	Strong gait vibrations amplified by the 3D-printed camera mount	Very high depth noise; fixed camera-to-base transform not valid during motion
(4) Reflective surfaces	Highly reflective floors and other surfaces in the building	Critical noise and errors in the active stereo depth

The second category concerns data recording (limitation 2). Because of the limited onboard storage, the recordings had to be written to an external USB storage device (Section 4.1.2). The robot, however, only provides external USB 2.0 ports, so that the camera, although capable of USB 3 operation, was also connected via USB 2.0. At 640 × 480 pixels, the D455 offers only the discrete frame rates of 5, 15 and 30 fps. Even at 15 fps, the available bandwidth was insufficient in early recordings: the delivered rates remained at approximately 11.7 Hz for colour and 10.7 Hz for depth, and frames were lost. Although a later recording reached an average rate close to 15 Hz, the behaviour varied between runs, so that complete and temporally consistent recordings at a usable frame rate could not be obtained reliably.

The third category concerns the quality of the depth data. The legged gait of the Go1 produces strong vibrations, which the 3D-printed camera mount amplified rather than damped. This resulted in very high noise in the depth measurements and, at the same time, meant that the fixed, manually defined camera-to-base transform did not hold while the robot was moving (limitation 3). Furthermore, the floors of the university building are highly reflective, and reflections on the floor and on other surfaces caused critical noise and errors in the depth measured by the active stereo sensor of the D455 (limitation 4).

The depth-quality limitations were decisive, since the framework depends on metric depth at every stage of the per-frame path. Depth is used to back-project segmentation masks into 3D observations (Section 3.3.1), it provides the 3D evidence on which data association is based (Section 3.3.2), and it is integrated by Hydra into the volumetric reconstruction from which the mesh, the object nodes and the places are derived [7]. In addition, back-projection relies on the rigid camera transform of Equation (3.1), which limitation 3 invalidates. Noisy depth therefore propagates into the object geometry, into the tracking and into the map itself. Together with the incomplete recordings, this made the Go1 data unsuitable as the basis of the evaluation.

4.1.4. Substitute Datasets

The substitute datasets were selected according to the input contract defined in Section 3.2. A suitable dataset had to provide registered colour and depth images, the camera calibration, odometry and a known extrinsic transform between the robot base and the camera, recorded in indoor environments as replayable ROS bags. Two datasets were chosen on this basis.

The office sequences of the OpenLORIS-Scene dataset [12] serve as the real-world data. They were recorded with a RealSense D400-series camera, that is, a sensor of the same family as the D455, and provide depth images that are already aligned to the colour image, real wheel odometry, and a calibrated transform from the robot base to the camera. OpenLORIS-Scene is therefore the closest real-world substitute for the planned recordings. Since the data were recorded on a wheeled robot with complete sensor streams, they avoid precisely the recording and vibration problems described in Section 4.1.3, so that the method can be evaluated on real sensor data without the results being dominated by these hardware-specific effects. The bags were converted into the ROS 2 format with the same procedure that had been planned for the Go1 data, and the 16-bit depth images in millimetres are converted into metres by the preprocessor.

The simulated office and apartment scenes of the uHumans2 dataset [11] complement the real-world data. Both provide ground-truth odometry and metric depth. The office scene was used for development, optimisation and repeatable experiments under controlled conditions, while the apartment scene, which contains a human, serves to test the portability of the framework to a second environment and its handling of a dynamic object.

5. Experiments

6. Conclusions

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A. Example appendix chapter

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